# Curs 1

Experiment → tipar → model teoretic

Mărimi fizice → cuantificarea utilizată pentru a descrie un fenomen

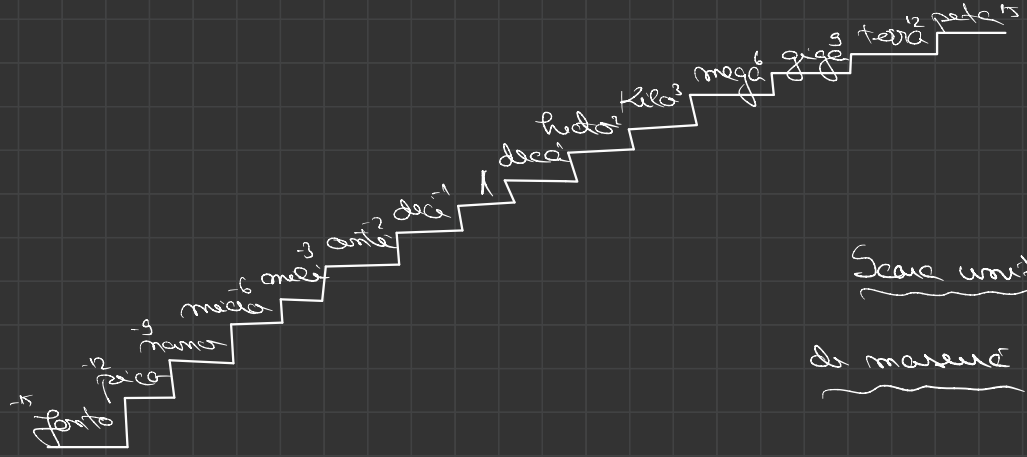
M = valoare numerică < unitate de măsură >

Ex: T = 300K

## Clasificarea mărimilor fizice

1. → scalare (doar val. numerică și unitate de măsură)
   → vectoriale (are și orientare spațială: direcție și sens)

2. → fundamentale (masă, timp, lungime, temp., int. curent, int. luminoasă, cantitatea de substanță)
   → derivate (ex: $v = \frac{d}{t}$ < m/s >)




Seare unitetilor de masure

| | |
|---|---|
| nucleul atomic | $10^{-14}$ m |
| atom | $10^{-10}$ m (Angstrom) |
| unei celule | $10^{-6}$ m |
| înălțimea omului | 1,7 m |
| diam. Pământ | $10^{7}$ m |
| dist Pământ - Soare | $10^{11}$ m |

# Aplicatii

Energia de repaus a unei particule $E = mc^2$

Energia de repaus a electronului

$m = 9,11 \cdot 10^{-31}$ kg

$c = 3 \cdot 10^8$ m/s

$\Rightarrow E = mc^2 = 9,11 \cdot 10^{-31} \cdot 3 \cdot 10^8 = 81,99 \cdot 10^{-15}$ J

$1eV = 1,6 \cdot 10^{-19}$ J

$51,24 \cdot 10^{11}$ eV

Marimi scalare -> se supun regulilor aritmetice simple

Marimile vectoriale -> adunare compunere

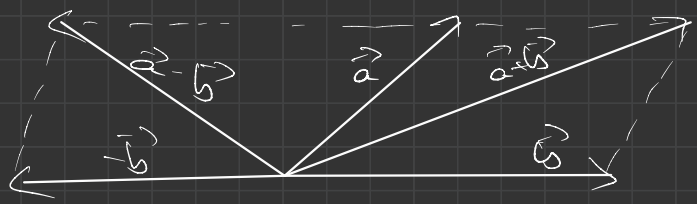
scadere descompunere

inmultire -> produs cu un scalar

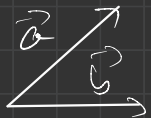
-> produs vectorial

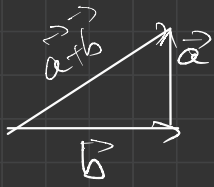
# Compunerea vectorilor

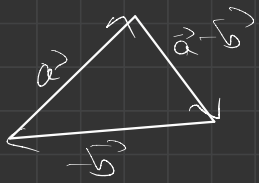
## Regula paralelogramului

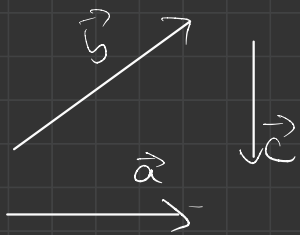



## Regula contur poligonal



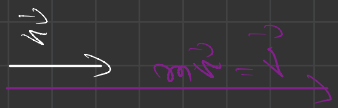



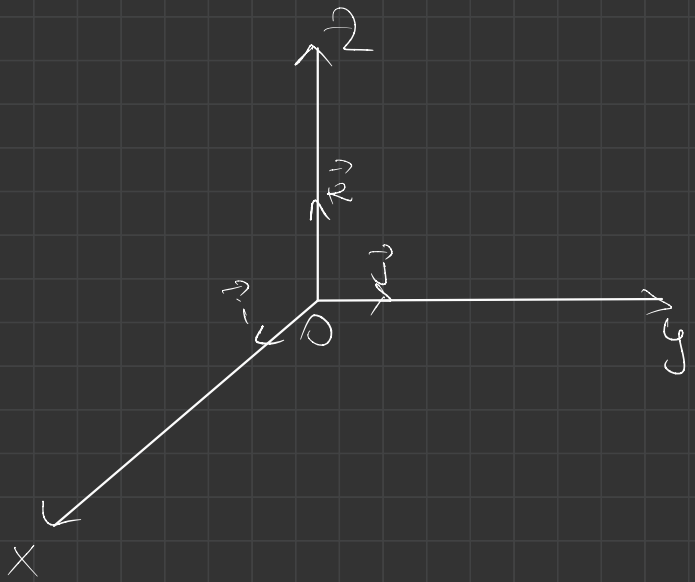



# Produsul cu un scalar

$\vec{v} \to \vec{v} = m \cdot \vec{u}$ , $m$ - număr scalar



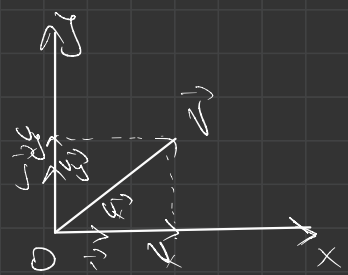

Aplicații: reprezentarea vectorului pe axe




$|\vec{i}| = |\vec{j}| = |\vec{k}| = 1$

$|\vec{i}| = |\vec{j}| = |\vec{k}| = 1$

2D:

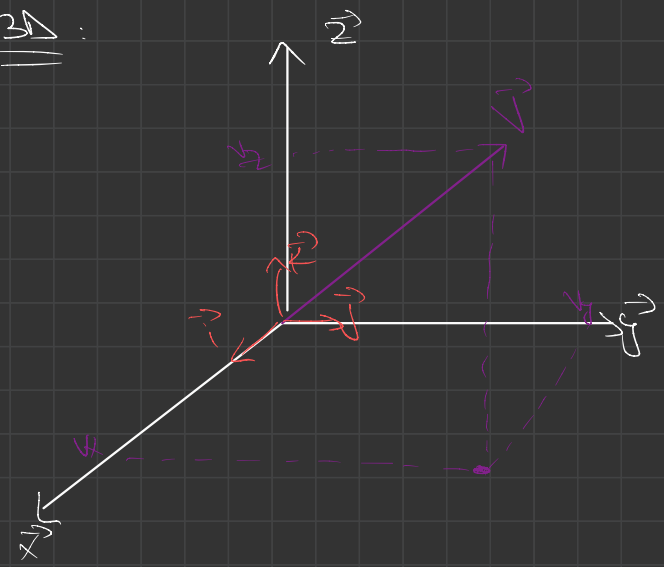



$\vec{v} = \vec{v_x} + \vec{v_y}$

$\vec{v_x} = x \cdot \vec{i}$

$\vec{v_y} = y \cdot \vec{j}$

3D:




$\vec{a}$ $\vec{b}$

$$\vec{a} = a_x \vec{i} + a_y \vec{j} + a_z \vec{k}$$

$$\vec{b} = b_x \vec{i} + b_y \vec{j} + b_z \vec{k}$$

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$$\vec{a} + \vec{b} = (a_x + b_x)\vec{i} + (a_y + b_y)\vec{j} + (a_z + b_z)\vec{k}$$

PRODUS SCALAR : $\vec{a} \cdot \vec{b} = a \cdot b \cdot \cos\alpha$

ex: lucrul mecanic

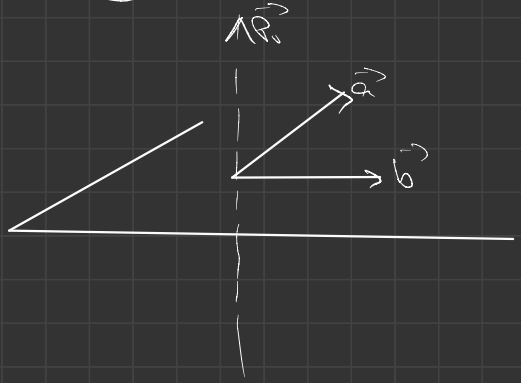
$$L = \vec{F} \cdot \vec{d}$$

$$L = F \cdot d \cdot \cos\alpha$$

$$\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = \vec{k} \cdot \vec{k} = 1$$

$$\vec{i} \cdot \vec{j} = \ldots = \vec{k} \cdot \vec{i} = 0$$

PRODUS VECTORIAL · $\vec{P_v} = \vec{a} \times \vec{b}$




$P_v = a \cdot b \cdot \sin\alpha$

$\alpha = \sphericalangle(\vec{a}, \vec{b})$

$$\vec{P_v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \vec{i}(a_y b_z - b_y a_z) + \vec{j}(b_x a_z - a_x b_z) + \vec{k}(a_x b_y - a_y b_x)$$

Operatorul nabla

$$\nabla = \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}$$

$$\vec{v} = v_x \vec{i} + v_y \vec{j} + v_z \cdot \vec{k}$$

$$\nabla \vec{v} = \vec{i}\vec{i}\frac{\partial v_x}{\partial x} + \vec{j}\vec{j}\frac{\partial v_y}{\partial y} + \vec{k}\cdot\vec{k}\frac{\partial v_z}{\partial z}$$

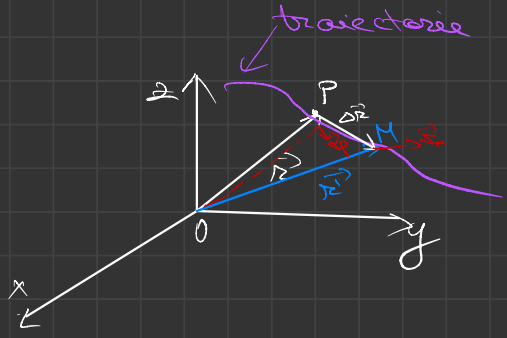
(divergent)

$$\nabla E = \hat{i}\frac{\partial E}{\partial x} + \hat{j}\frac{\partial E}{\partial y} + \hat{k}\frac{\partial E}{\partial z} \quad \text{(gradient)}$$

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} \quad \text{(rotor)}$$

# Curs 2

## CINEMATICĂ → se ocupă cu studierea mișcării unui corp (fără rezultate)




$\vec{r}$ = vectorul de poziție

$\vec{r}$ = vectorul poziție

$\vec{r} = f(t)$ legea de mișcare

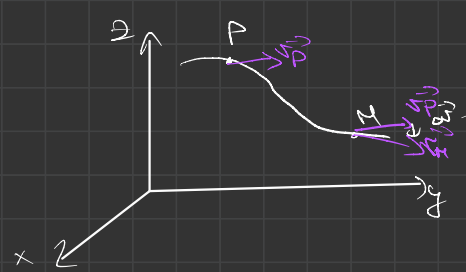
**Deplasarea** : diferența dintre 2 vectori : $\Delta\vec{r} = \vec{r} - \vec{r}$

**Viteza** : $\vec{v} = \frac{\Delta\vec{r}}{\Delta t}$ (viteza medie)

$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} = \dot{\vec{r}}$ (derivata) (viteza instantanee)

(v are același sens cu dr)

**Accelerația** : $\vec{a} = \frac{\Delta\vec{v}}{\Delta t}$ (accelerația medie)

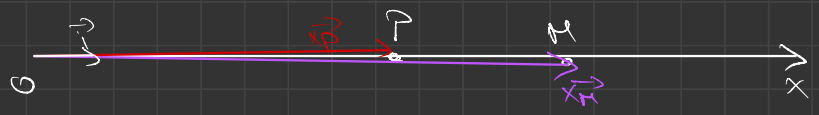



→ diferența dintre v din punctul P și cel din M

$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \dot{\vec{v}}$ (derivata)

$\vec{v} = \frac{d\vec{r}}{dt}$ $\Rightarrow$ $\vec{a} = \frac{d^2\vec{r}}{dt^2} = \ddot{\vec{r}}$ (derivata de ord 2)

## MIȘCAREA UNIDIMENSIONALĂ




$\vec{r} = \vec{i}x + \vec{j}y + \vec{k}z$ $\Rightarrow$ $\vec{r} = \vec{i}x$

$y = 0$, $z = 0$

$\vec{v}_m = \frac{\Delta \vec{r}}{\Delta t}$ $\quad v = \dot{x}$ $\quad$ $\vec{a} = \ddot{x}\vec{i}$

Ex1 Legea de mișcare a unui mobil este dată de:

$x(t) = \alpha t + \frac{\beta t^3}{3}$, unde $\alpha = 3\ m/s$

$\beta = 0,1\ m/s^3$.

Calculați: $v_m = ?$ $t \in [0,5]$

$a_m = ?$ $t \in [0,5]$

$v = ?$ $a = ?$ $t = 5s$

$x(0) = 0\ m$

$x(5) = 3 \cdot 5 + 0,1 \cdot \frac{5^3}{3} = 15 + 4,16 = 19,16\ m$

$v_m = \frac{x(5) - x(0)}{5} = \frac{19,16}{5} = 3,83\ m/s$

$v(t) = \alpha + \beta t^2$ (derivată legii)

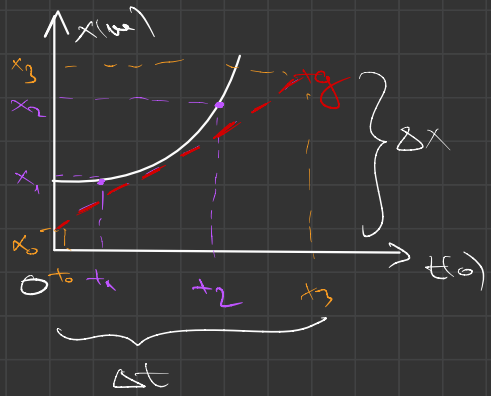
$v(0) = 3\ m/s$

$v(5) = 3 + 0,1 \cdot 25 = 5,5\ m/s$

$a_m = \frac{v(5) - v(0)}{5} = \frac{2,5}{5} = 0,5\ m/s^2$

$a(t) = 2\beta t$ (mai derivăm o dată legea)

$a(5) = 2 \cdot 0,1 \cdot 5 = 1\ m/s^2$



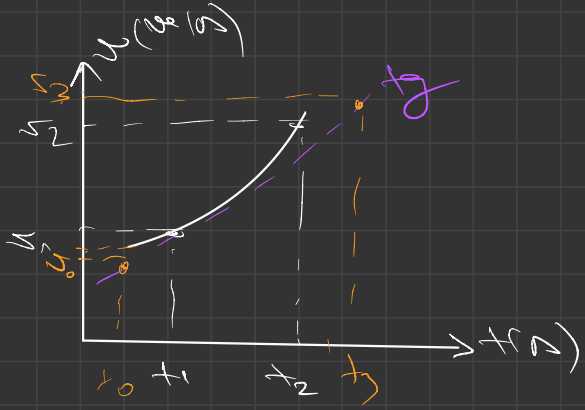

$$v_m = \frac{x_2 - x_1}{t_2 - t_1}$$

$f(x) = ax + b$

reale $\left( t_g \frac{x}{t} \left( = a = \text{pentru} \right) \right)$

fizică $t_g \frac{\Delta x}{\Delta t}$ (Δx nu se încearcă în u.m.)

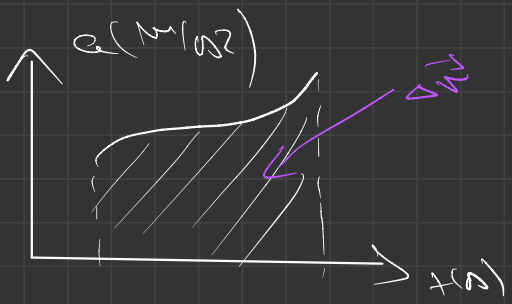
$$v = \frac{\Delta x}{\Delta t} = \frac{x_3 - x_0}{t_3 - t_0} \ m/s$$




$$a_m = \frac{v_2 - v_1}{t_2 - t_1}$$

$$a = \frac{\Delta v}{\Delta t} = \frac{v_3 - v_0}{t_3 - t_0}$$

(Derivata 0.3 sunt toate pe tangente)




$$\Delta v = \int_{t_1}^{t_2} a \, dt \quad \Rightarrow$$

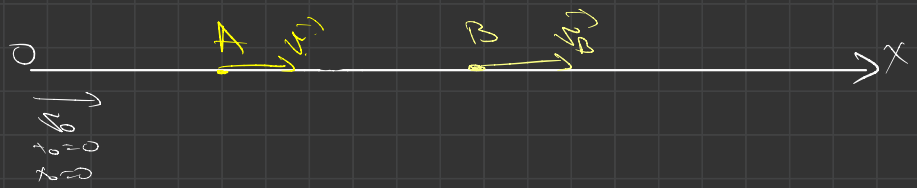
$$\int_{v_1}^{v_2} dv = \int_{t_1}^{t_2} a \, dt$$

$t_1 = 0 \qquad v(t) = \int_0^t a \, dt$

$$a = \frac{dv}{dt} \Rightarrow \int a \, dt = \int dv$$

$$v = \frac{dx}{dt} \Rightarrow \int v \, dt = \int dx$$

# MIȘCAREA RECTILINIE UNIFORM VARIATĂ ($a = ct$)




$$\int_{v_0}^{v} dv = \int_{0}^{t} a\,dt \quad \Rightarrow \quad v - v_0 = at$$

$$\Rightarrow \boxed{v = v_0 + at}$$

$$\int_{x_0=0}^{x} dx = \int_{0}^{t} v\,dt = \int_{0}^{t} (v_0 + at)\,dt$$

$$\Rightarrow \quad x - x_0 = v_0 t + \frac{a}{2} t^2$$

$$\Rightarrow \boxed{x = x_0 + v_0 t + \frac{at^2}{2}}$$

(Ex2) Accelerație unei motociclete e descrisă de legea:

$a(t) = At - Bt^2$, unde $A = 1{,}5\ m/s^3$

$B = 0{,}120\ m/s^2$

Motocicleta pornește din repaus la $t_0 = 0$, $v_0 = 0$.

Determinați: $x = f(t)$ legea mișcării

$v = f(x)$ legea vitezei

$v_{max} = ?$

$$\int_{v_0=0}^{v} dv = \int_0^t a\,dt$$

$$\Rightarrow v = \int At - Bt^2 = \frac{At^2}{2} - \frac{Bt^3}{3}$$ (v integrat)

$$\int_{x_0}^{x} dx = \int_0^t v\,dt = \int_0^t \left(\frac{At^2}{2} - \frac{Bt^3}{3}\right) dt$$

$$\Rightarrow x - x_0 = \frac{At^3}{6} - \frac{Bt^4}{12}$$ (v integrat a doua oară)

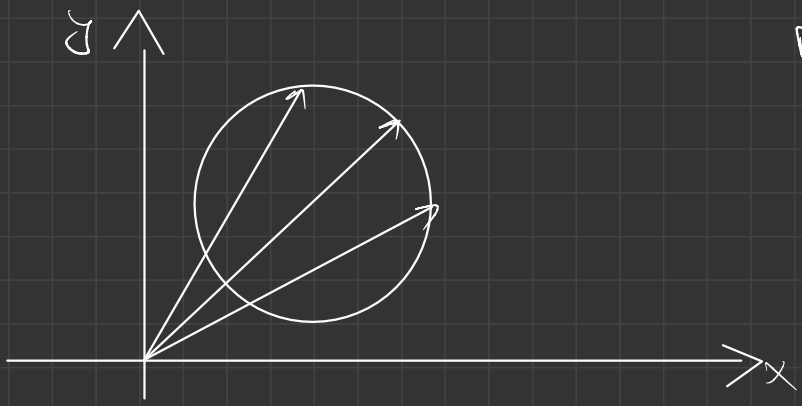
$x_0 = 0$

$$\frac{dy}{dt} = 0 \quad \Rightarrow \quad At - Bt^2 = 0$$

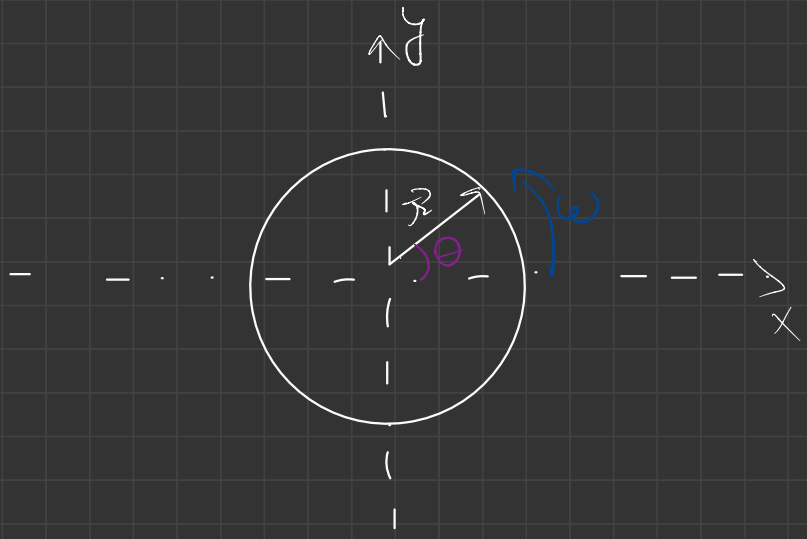
$$\Rightarrow \quad t = \frac{A}{B} \qquad \left( \text{Pr } t = \frac{A}{B} \text{ obs} \right)$$

$$y\left(\frac{A}{B}\right) = y_{max} = \frac{A}{2}\left(\frac{A}{B}\right)^2 - \frac{B}{3}\left(\frac{A}{B}\right)^3$$

# MIȘCAREA CIRCULARĂ




$R = f(t)$




$(x, y) \rightarrow (r, \theta)$ (trecem la sistemul polar)

$R(t) \rightarrow \theta = f(t)$

$\omega$ - viteza unghiulară

$$\omega = \frac{\Delta\theta}{\Delta t}$$

$\theta = 2\pi \Rightarrow$ timpul total în perioadă $\overset{\text{not}}{=} T = \Delta t$

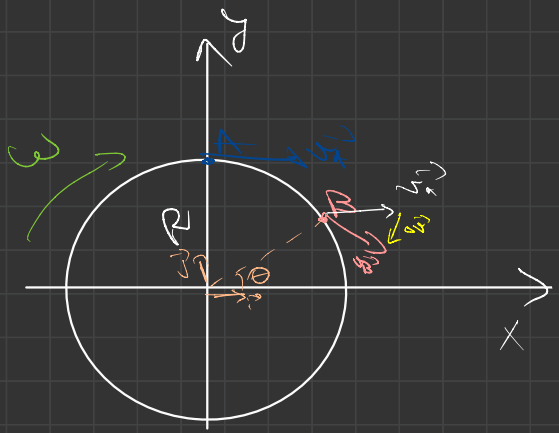
$$\int_0^T \omega \, dt = \int_0^{2\pi} d\theta \quad \Rightarrow \quad \omega T = 2\pi$$

$$\Rightarrow \boxed{T = \frac{2\pi}{\omega}}$$ perioada mișcării

$\nu = \frac{1}{T}$ (frecvență) → câte rotații sunt în intervalul de timp

$[\nu]_{SI} = 1\,Hz = 1\,s^{-1}$

## MIȘCARE CIRCULARĂ UNIFORMĂ




$\vec{a}_m = \frac{\Delta \vec{v}}{\Delta t}$, $\Delta \vec{v} = \vec{v}_B - \vec{v}_A$

$v = v_A = v_B$

$\vec{v}_A = v\,\vec{i}$

$\vec{v}_B = v \sin\theta\,\vec{i} - v\cos\theta\,\vec{j}$

sensul spre $\vec{v}$

$\vec{v}_B - \vec{v}_A = \vec{i}(v\sin\theta - v) - \vec{j}(v\cos\theta) \neq 0$

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \vec{v}_x \cdot a_x + \vec{v}_y \cdot a_y$$

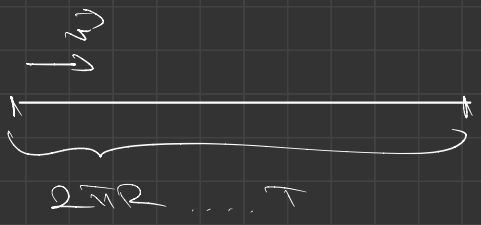
$$|\vec{r}_1| = |\vec{r}_2| = R$$

$$|\vec{v}_1| = |\vec{v}_2| = v$$

$\triangle OAB \sim \triangle$ скорости

$$\frac{|\Delta \vec{v}|}{|\Delta \vec{r}|} = \frac{v}{R} \Rightarrow \frac{|\Delta \vec{v}|}{v \cdot \Delta t} = \frac{v}{R} \Rightarrow \boxed{a = \frac{v^2}{R}}$$

А т. В близко $\Rightarrow AB = |\Delta \vec{r}| = v \cdot \Delta t$

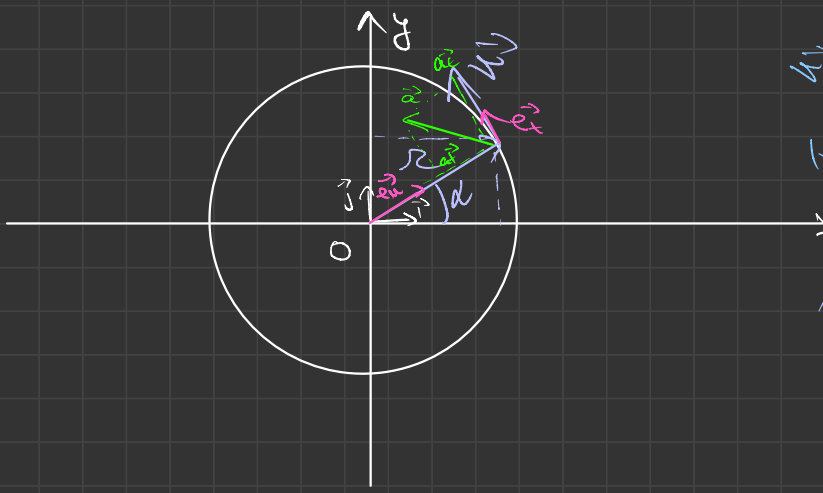



$2\pi R = vT \quad \Rightarrow v = \frac{2\pi R}{T}$

$$a = \frac{4\pi^2 R}{T^2}$$

# Curs 3

## MIȘCAREA CIRCULARĂ NEUNIFORMĂ




$\vec{v} \neq const$

$|\vec{v}| \neq const$

$r_x = r\cos\alpha$

$r_y = r\sin\alpha$

$$\Rightarrow \boxed{\vec{r} = \vec{i} \cdot r\cos\alpha + \vec{j} \cdot r\sin\alpha}$$

$$\boxed{\vec{v} = \frac{d\vec{r}}{dt} = (-\vec{i} \cdot r\sin\alpha + \vec{j} \cdot r\cos\alpha)\dot{\alpha}}$$

$\dot{\alpha}$ derivata lui $\alpha$ în raport cu timpul $\left(\frac{d\alpha}{dt}\right)$

$\vec{e}_n = \vec{i}\cos\alpha + \vec{j}\sin\alpha$

$\vec{e}_t = (-\vec{i}\sin\alpha + \vec{j}\cos\alpha)$

$\vec{r} = r \cdot \vec{e}_n$

$\vec{v} = r \cdot \dot{\alpha} \cdot \vec{e}_t$

$$\vec{a} = \frac{d}{dt}\left(\dot{\alpha}\left(-\vec{i}\, r \sin\alpha + \vec{j}\, r\cos\alpha\right)\right) =$$

$$= \ddot{\alpha} r\left(\underbrace{-\vec{i}\sin\alpha + \vec{j}\cos\alpha}_{\vec{e}_T}\right) + \dot{\alpha}^2 r\left(\underbrace{-\vec{i}\sin\alpha + \vec{j}\cos\alpha}_{-\vec{e}_N}\right)$$

$$= \boxed{\ddot{\alpha} r\, \vec{e}_T - \dot{\alpha}^2 r\, \vec{e}_N}$$

$$v = \text{const} = r\dot{\alpha} = \text{const}$$

$$\dot{\alpha} = \text{const} \Rightarrow \ddot{\alpha} = \text{const}$$

$$\Rightarrow \boxed{a_T = a_N}$$

$$\dot{\alpha} = \omega \quad \text{(viteză unghiulară)}$$

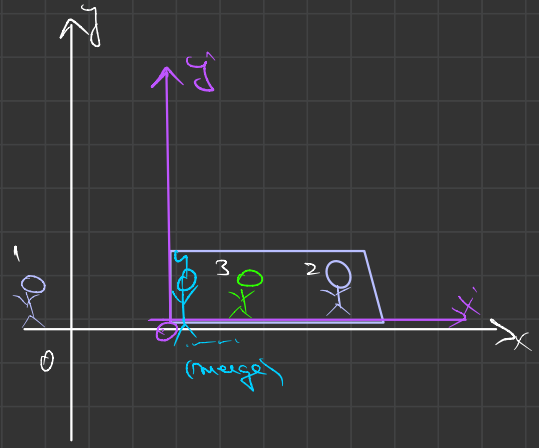
$$\Rightarrow \boxed{a_N = \omega^2 r}$$

Legile de mișcare

$$\text{MRU} \begin{cases} x = f(t) \Rightarrow v_x(t) \Rightarrow a_x(t) \\ y = f(t) \Rightarrow v_y(t) \Rightarrow a_y(t) \end{cases}$$

$$\text{MCU} \begin{cases} \alpha = f(t) \Rightarrow \omega = f(t) \Rightarrow v(t) \Rightarrow a(t) \end{cases}$$

# Reactivitatea în mecanica clasică




1 → 2 se deplasează cu $v_0$

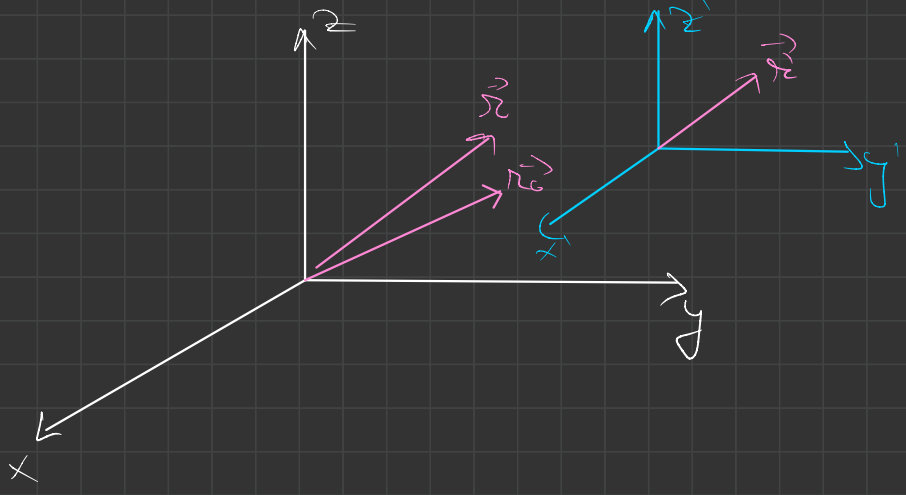
2 → 3 stă în mașină

2 → 4 se deplasează cu $-v_0$

1 → 4 se deplasează cu $v_0 + (-v_0)$

$\vec{v}_r = \vec{v} - \vec{v}_0$ viteza relativă a lui 4 față de Oxy

Cazul 3D




$\vec{r} = \vec{r}_0 + \vec{r}'$

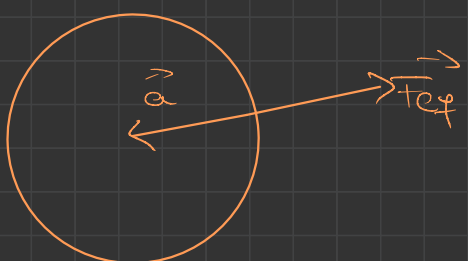
$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d\vec{r}_0}{dt} + \frac{d\vec{r}'}{dt}$$

$$\vec{v} = \vec{v}_0 + \vec{v}'$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d\vec{v}_0}{dt} + \frac{d\vec{v}'}{dt}$$

## Relativitatea la rotație

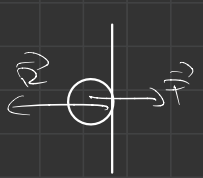
## Forța Coriolis

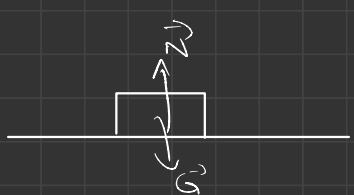



## Principiile mecanicii clasice

① $\vec{F}_{12} \longrightarrow \vec{R}' = -\vec{F}_{12}$ (acțiune – reacțiune)

(egale în modul, au de-a lungul aceeași dreaptă)



② $\vec{F} = m \cdot \vec{a}$ (principiul fundamental)

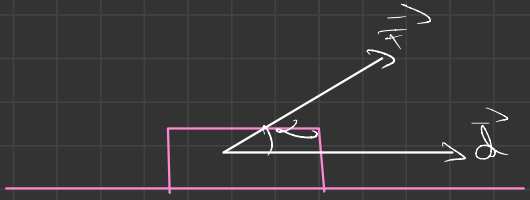
③ $\sum \vec{F} = \vec{R}_{rez} \Rightarrow \sum m \cdot \vec{a} = m \cdot \vec{a}_{rez}$

$\sum \vec{a} = \vec{a}_{rez}$

④ $\sum \vec{F} = 0 \quad \vec{v} = \text{const}$ (ie mișcare, direcție și sens)

(principiul inerției)

## Lucrul mecanic




$$L = \vec{F} \cdot \vec{d} = F \cdot d \cdot \cos\alpha$$

$\langle L \rangle_{Si} = J$

$\delta L$ = lucru mecanic elementar

$\delta L = \vec{F} \cdot \delta \vec{r} \Rightarrow L = \int_{\vec{r}_1}^{\vec{r}_2} \delta L = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} d\vec{r}$

$$L = \int_{x_1}^{x_2} \vec{F} \cdot d\vec{x} = \int_{x_1}^{x_2} m \vec{a} \, d\vec{x} =$$

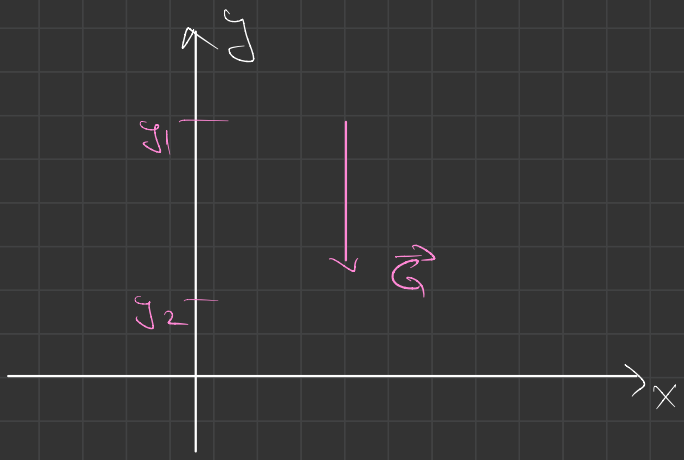
$$= \int \frac{m d\vec{v}}{dt} d\vec{x} = \int_{v_1}^{v_2} m \vec{v} \, d\vec{v} = \frac{m v_2^2}{2} - \frac{m v_1^2}{2}$$

$$\Delta E_c = \frac{m v^2}{2}$$

$$L = \Delta E_c$$

$$\vec{F} = m \cdot \vec{a} = \frac{m d\vec{v}}{dt} = \frac{d(m\vec{v})}{dt} = \frac{d\vec{p}}{dt}$$ (se puede)

# Energia potențială gravitațională



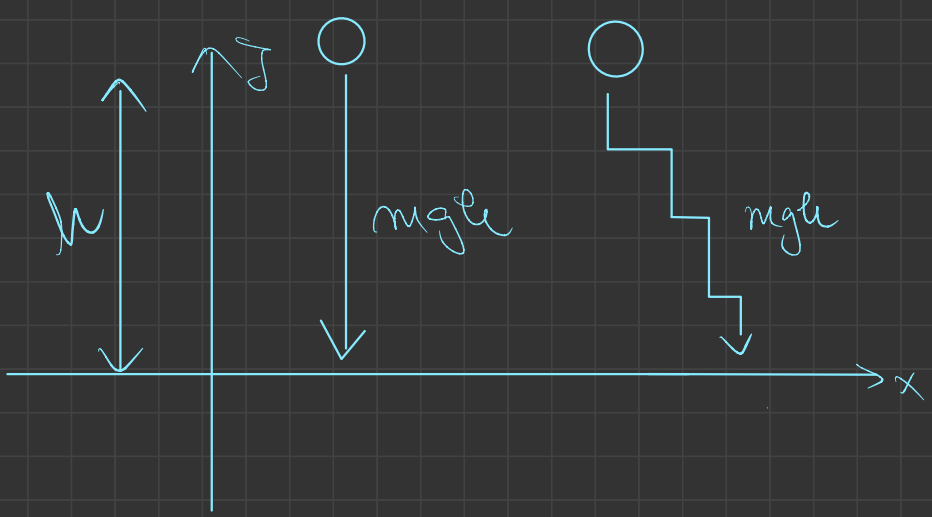

$$L_G = \vec{G}(\vec{r}_2 - \vec{r}_1) = \vec{G}\cdot\vec{d}$$

$$= -mg\vec{j}(\vec{r}_{y_1} - \vec{r}_{y_2}) =$$

$$= -(mgy_2 - mgy_1)$$

$$\Rightarrow L_G = -(E_{p2} - E_{p1})$$

$$\boxed{L_G = -\Delta E_p}$$




Lucrul mecanic e independent de traiectorie ⇒ Conservativă

# Conservazione energia

$\vec{R} = \vec{F_C} + \vec{F_N}$

$\vec{F_C}$: forza conservativa

$\vec{F_N}$: forza non conservativa

$$L_R = L_{F_C} + L_{F_N} \Rightarrow \Delta E_c + \Delta E_p = L_{F_N}$$

$$\Delta(E_c + E_p) = L_{F_N}$$

$E_T$

$$L_R = \Delta E_c$$

$$L_{F_C} = -\Delta E_p$$

$$\Delta E_T = L_{F_N}$$

# Energia potenziale

$$\delta L = \vec{F} \, d\vec{s}$$

$$\vec{F} = \vec{i} F_x + \vec{j} F_y + \vec{k} F_z$$

$$d\vec{s} = \vec{i} dx + \vec{j} dy + \vec{k} dz$$

$$\Rightarrow F_x dx + F_y dy + F_z dz = \delta L$$

$$\delta L = -\delta E_p =$$

$$= -\left[\left(\frac{\partial E_p}{\partial x}\right)dx + \left(\frac{\partial E_p}{\partial y}\right)dy + \left(\frac{\partial E_p}{\partial z}\right)dz\right]$$

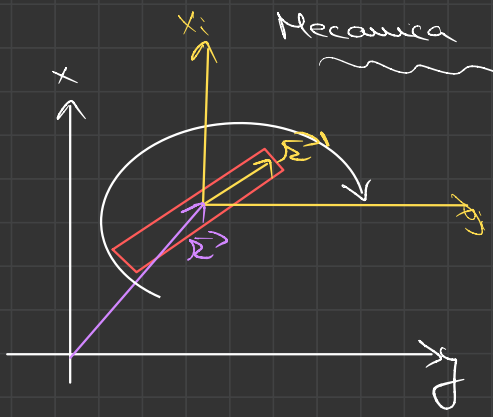
$$\Rightarrow \begin{cases} F_x = -\dfrac{\partial E_p}{\partial x} \\ F_y = -\dfrac{\partial E_p}{\partial y} \\ F_z = -\dfrac{\partial E_p}{\partial z} \end{cases}$$

$$\vec{F} = -\left(\vec{i}\,\frac{\partial E_p}{\partial x} + \vec{j}\,\frac{\partial E_p}{\partial y} + \vec{k}\,\frac{\partial E_p}{\partial z}\right) = -\nabla E_p \quad \text{(gradient)}$$

$$\Rightarrow \boxed{\vec{F} = -\nabla E_p}$$

# Curs 4

## Mecanica solidului rigid

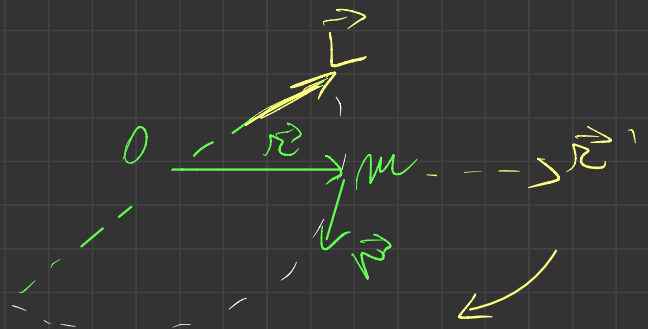


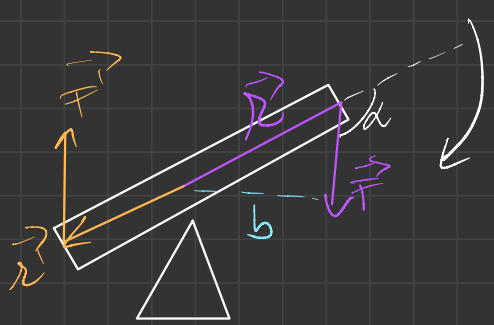

$\Sigma = const \Rightarrow$ repaus?

(se poate roti)

### Momentul cinetic

$$\vec{L} = \vec{r} \times \vec{p}$$




$$\vec{M} = \vec{r} \times \vec{F}$$

$$M = r \cdot F \cdot \sin\alpha = F \cdot b$$

$$\sin(180^\circ - \alpha) = \frac{b}{r}$$

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r}\times\vec{p}) = \frac{d\vec{r}}{dt}\times\vec{p} + \vec{r}\times\frac{d\vec{p}}{dt}$$

$\frac{d\vec{r}}{dt}\times\vec{p} = \vec{v}\times m\vec{v} = 0$

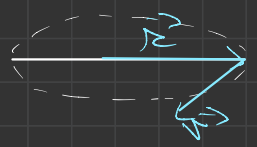
$\vec{r}\times\frac{d\vec{p}}{dt} = \vec{r}\times m\frac{d\vec{v}}{dt} = \vec{r}\times\vec{F}$

$$\Rightarrow \boxed{\frac{d\vec{L}}{dt} = \vec{r}\times\vec{F} = \vec{M}}$$

Théorème variation du moment cinétique

Moment est nul $\Rightarrow$ mom cinétique constant

$\vec{L} = \vec{r}\times\vec{p}$

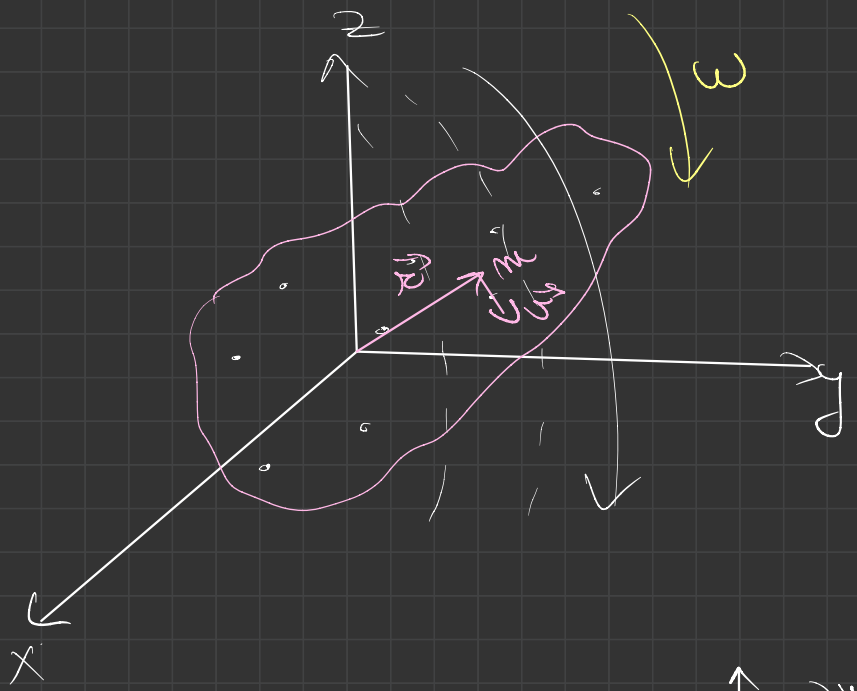



$\alpha(\vec{r},\vec{v}) = 90°$

$\Rightarrow L = mvr = \text{const}$

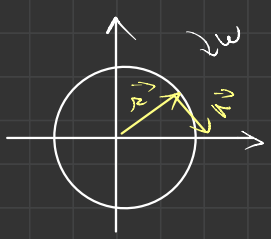
$mr_1v_1 = mr_2v_2$

# Energía cinética




$$E_c = \frac{m v^2}{2}$$

$$E_t = \sum_i \frac{m_i v_i^2}{2}$$




$$\omega^2 r = \frac{v^2}{r} \quad \forall \quad v^2 = \omega^2 r^2$$

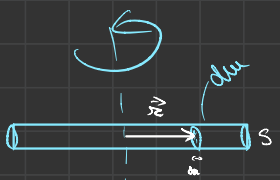
$$\omega = \dot{\theta}$$

$$\Rightarrow \forall \quad E_t = \sum_i \frac{m_i r_i^2 \omega^2}{2} \quad \forall$$

$$\boxed{E_t = \frac{I \omega^2}{2}}$$

$$\Rightarrow \boxed{I = \sum_i m_i r_i^2}$$

# Exemple de calcul pentru momentul de inerție

① 


$$I = \int dm\, x^2 = \int x^2 dm$$

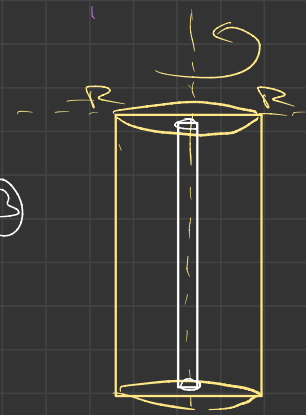
$$dm = \int dV = \int \rho S\, dx$$

$$\Rightarrow I = \int_{-L/2}^{L/2} \rho S x^2\, dx = \rho S \frac{x^3}{3}\Big|_{-L/2}^{L/2}$$

$$I = \frac{\rho S}{3}\left(\frac{L^3}{8} + \frac{L^3}{8}\right) = \frac{\rho S L^3}{12} = \frac{ML^2}{12} \qquad (M = \rho S L)$$

②

$$I = \int_0^L \rho S x^2\, dx = \frac{\rho S L^3}{3} = \frac{ML^2}{3}$$

③ 


$M, L, \Delta$

$$I = \int x^2\, dm = \int_0^R 2\pi \rho L x^3\, dx$$

$$V = \pi x^2 L$$

$$dV = 2\pi x L\, dx$$

$$\Rightarrow dm = \rho dV = \rho 2\pi x L$$

$$\Rightarrow I = 2\pi \rho L \frac{R^4}{4} = \frac{MR^2}{2}$$

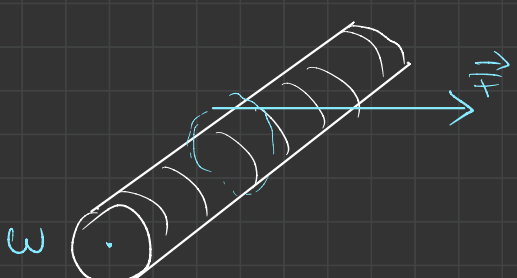
(Ex) Un fir inextensibil cu masa neglijabilă este înfășurat de mai multe ori pe un cilindru solid.

$m_{cilindru} = 50\,kg$

$d_{cilindru} = 0{,}12\,m$. Cilindrul se poate roti în jurul unei axe fixe orizontale de o frecare. Capătul liber al firului este tras cu $F_{exert} = 9\,N$ pe o distanță de 2 m. Calculați:

a) $\omega = ?$ (viteza unghiulară)

b) $v = ?$ (viteza firului după o parcurs 2m)




$L = \Delta E_c = 18\,J$

$\Delta E_c = E_{cf} - E_{ci}$

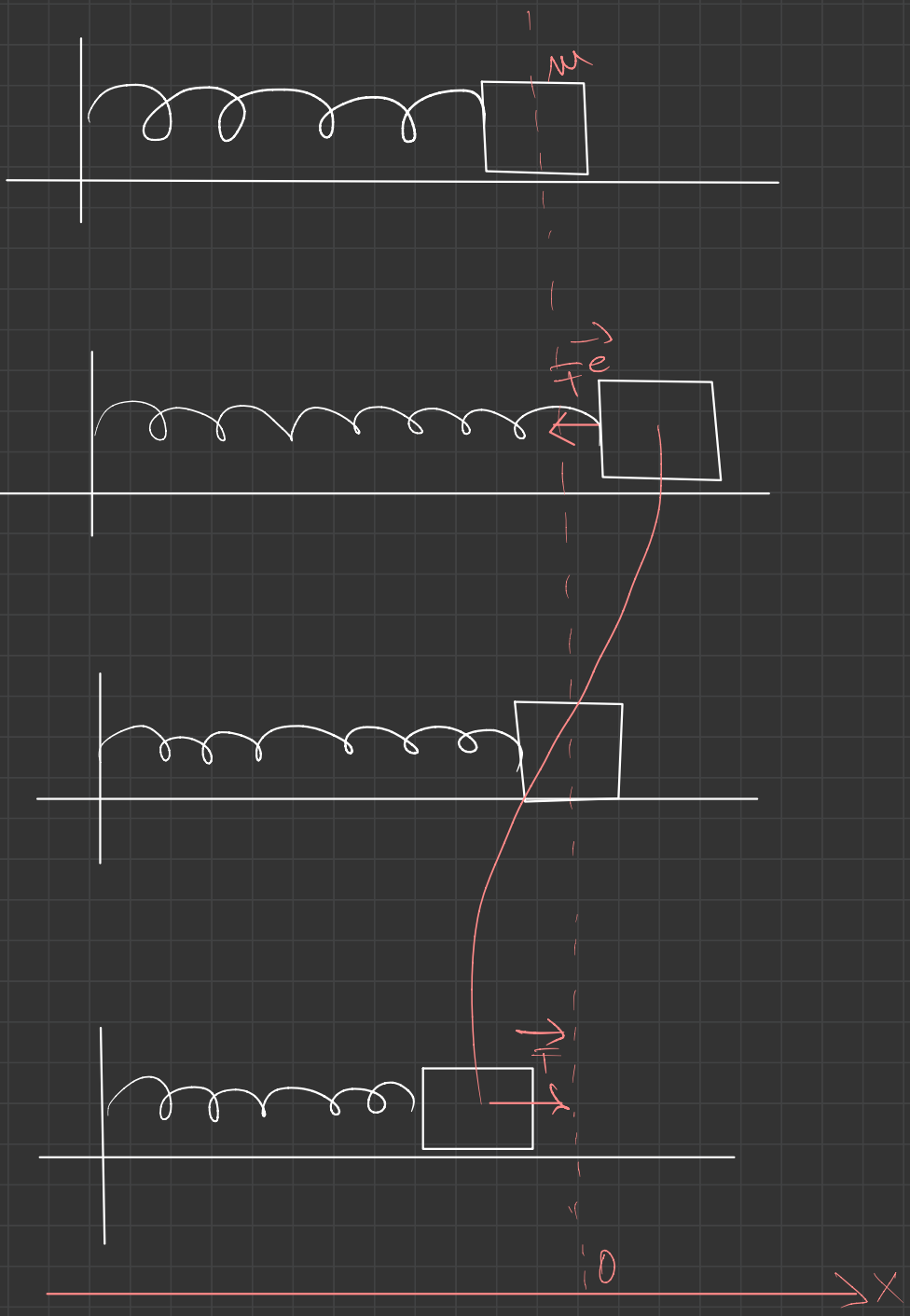
$E_{ci} = 0$

$E_c = \frac{I\omega^2}{2} = \frac{MR^2}{2}\cdot\frac{\omega^2}{2}$

$\Rightarrow \omega^2 = \frac{4Fd}{MR^2} = 400 \Rightarrow \omega = 20\ rad/s$

$v = \omega R = 20 \cdot 0{,}06 = 1{,}2\ m/s$

# Mezcla oscilatoria



$$\vec{F} = -k \cdot \vec{x}$$

$$F = -kx$$

$$F = m \cdot a = m \cdot \frac{dv}{dt} = m \cdot \frac{d^2x}{dt^2}$$

$$\frac{m\, d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} + \frac{kx}{m} = 0$$

$$\boxed{\frac{k}{m} = \omega^2}$$

$$\ddot{x} + \omega^2 x = 0$$

$$x = e^{\alpha t}$$

$$\Rightarrow \alpha^2 e^{\alpha t} + \omega^2 e^{\alpha t} = 0 \Rightarrow \alpha^2 = -\omega^2$$

$$\alpha = \pm \omega i$$

$$x = A e^{i\omega t} + B e^{-i\omega t}$$

Euler:

$$e^{i\omega t} = \cos \omega t + i \sin \omega t$$

$$e^{-i\omega t} = \cos \omega t - i \sin \omega t$$

$$\Rightarrow x = (A+B) \cos \omega t + i (A-B) \sin \omega t$$

$$x = (A+B) \cos \omega t = x_{max} \cos \omega t$$

Caz general:

$$\boxed{x = x_{max} \cdot \cos(\omega t + \varphi)}$$

c- Faça la mon sinusoidal e obtenha la mon orbital ⇒

$$x = x_{max} \cos(\omega t + \varphi)$$

$$x = x_{max} \operatorname{sen}(\omega t + \varphi)$$

# Curs 5

## Oscilatorul armonic liniar

$\vec{F} = -k\vec{x} \Rightarrow$ **$x = A\cos(\omega t + \varphi)$**

$A = x_{max}$ (amplitudine)

$\omega = \frac{2\pi}{T} = 2\pi\nu$

$\varphi$ = faza inițială

viteza: $v(t) = \frac{dx}{dt} = -\omega A \sin(\omega t + \varphi)$

accelerația: $a(t) = -\omega^2 A \cos(\omega t + \varphi)$ (derivăm)

## Energia oscilatorului armonic

**$E = E_p + E_c$**

$E_c = \frac{mv^2}{2} = \frac{m\omega^2 A^2 \sin^2(\omega t + \varphi)}{2}$

$E_p = \frac{k}{2} A \cos^2(\omega t + \varphi)$

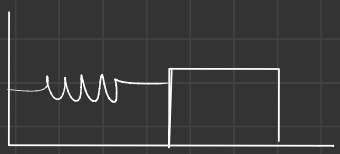
$\vec{F} = -\text{grad } E_p \Rightarrow Kx = -\frac{dE_p}{dx} \Rightarrow \int dE_p =$

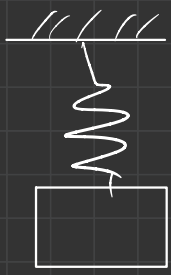
$= \int Kx\,dx = \frac{Kx^2}{2}$

$k = \omega^2 m \Rightarrow E_p = \frac{m\omega^2 A^2}{2} \cos^2(\omega t + \varphi)$

$\Rightarrow \boxed{E_c + E_p = \frac{m\omega^2 A^2}{2}}$ Energia totală

## Oscilații amortizate





$m\vec{a} = \sum_i \vec{F}_i$

$m\ddot{x} = -kx - b\dot{x} \Rightarrow (*)$

$F_e = -kx$

$F_R = -bv = -b\dot{x}$

$(*) \Rightarrow m\ddot{x} + b\dot{x} + kx = 0 \quad / m$

$\ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = 0$

$\frac{b}{m} = 2\beta$

$\frac{k}{m} = \omega^2$

$$\ddot{x} + 2\gamma\dot{x} + \omega^2 x = 0$$

$$x = e^{\alpha t} \longrightarrow \alpha^2 e^{\alpha t} + 2\gamma\alpha e^{\alpha t} + \omega^2 e^{\alpha t} = 0$$

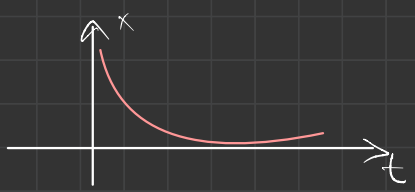
$$\alpha^2 + 2\gamma\alpha + \omega^2 = 0$$

$$\alpha = \frac{-2\gamma \pm 2\sqrt{\gamma^2 - \omega^2}}{2}$$

$$\boxed{\alpha = -\gamma \pm \sqrt{\gamma^2 - \omega^2}}$$

Caso 1 $\quad (\gamma > \omega) \Rightarrow \alpha \in \mathbb{R}$

$$x = C_1 e^{\left(-\gamma + \sqrt{\gamma^2-\omega^2}\right)t} + C_2 e^{\left(-\gamma - \sqrt{\gamma^2-\omega^2}\right)t}$$




(smorzamento aperiodico)

Caso 2 $\quad (\gamma = \omega) \Rightarrow \alpha \in -\gamma$

$$x = C_1 e^{-\gamma t}$$

Caso 3 $\quad (\gamma < \omega) \Rightarrow \alpha = -\gamma \pm \underbrace{\sqrt{\omega^2 - \gamma^2}\, i}_{\omega'}$

$$x = C_1 e^{\left(-\gamma + \sqrt{\omega^2-\gamma^2}\, i\right)t} + C_2 e^{\left(-\gamma - \sqrt{\omega^2-\gamma^2}\, i\right)t}$$

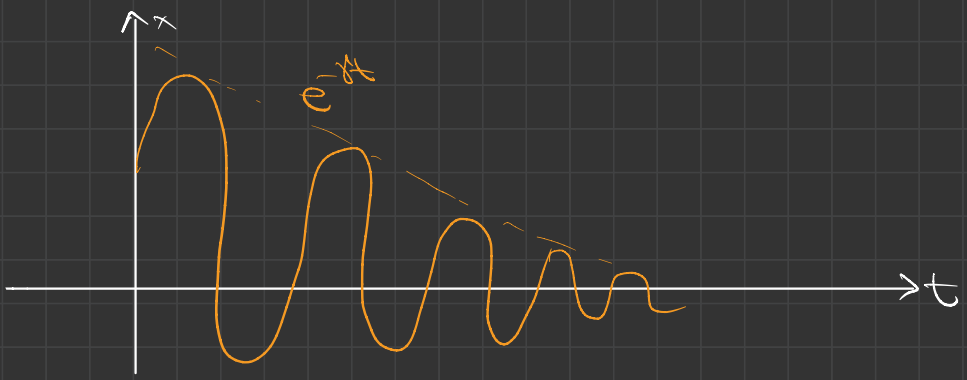
$$x = e^{-\gamma t}\left(C_1 e^{+t\sqrt{\omega^2-\gamma^2}\, i} + C_2 e^{-t\sqrt{\omega^2-\gamma^2}\, i}\right)$$

$$\Rightarrow x = e^{-\gamma t}\left(C_1 e^{+\omega' i} + C_2 e^{-\omega' i}\right) \Rightarrow \boxed{x = x_{max}\, e^{-\gamma t}\cos(\omega t + \varepsilon)}$$

$$\omega' = \sqrt{\omega^2 - \gamma^2}$$

$\omega$ = pulsația proprie

$\omega'$ = pulsația reală a oscilatorului




## Decrementul logaritmic

$$\delta = \frac{x(t)}{x(t+T)}, \qquad x = \text{elongația mișcării}$$

$T$ = perioada

$$= \frac{x_{max}\, e^{-\gamma t}\cos(\omega t + \varepsilon)}{x_{max}\, e^{-\gamma(t+T)}\cos(\omega(t+T) + \varepsilon)}$$

$$= e^{\gamma T} \Rightarrow \boxed{\ln \delta = \gamma T}$$

# Energia pierdută prin amortizare

$$\delta L_{F_r} = dE$$

$$\delta L_{F_r} = -F_r dx = -bv\,dx = -b\left(\frac{dx}{dt}\right)dx$$

Putere mecanică

$$P = \frac{dL}{dt} = -b\left(\frac{dx}{dt}\right)\left(\frac{dx}{dt}\right) = -bv^2 = -2\gamma\, mv^2$$

## Oscilații forțate

$$F_i = F_0 \cos(\omega' t)$$

$$m\cdot a = \sum F \quad \Rightarrow \quad m\ddot{x} = -kx - b\dot{x} + F_0\cos(\omega' t)$$

$$\ddot{x} + 2\gamma\dot{x} + \omega^2 x = F_0\cos(\omega' t)$$

$$x = x_0(t) + x_1(t)$$

$x_0$ – sol. ec. omogene

$x_1$ – sol ec. neomogene particulară

$$t \to \infty \quad \Rightarrow \quad x_0(t) \to 0 \quad \Rightarrow x = x_1(t) \to A\cos(\omega t + \varphi)$$

$= x_1(t)$

$$\ddot{x}_1(t) + 2\gamma \dot{x}_1(t) + \omega^2 x_1 = \frac{F_0}{m} \cos(\omega' t)$$

$$-A\omega'^2 \cos(\omega' t + \varepsilon) - 2\gamma \omega' A \sin(\omega' t + \varepsilon) +$$

$$+ \omega^2 A \cos(\omega' t + \varepsilon) = \frac{F_0}{m} \cos(\omega' t)$$

$$-A\omega'^2 (\cos\omega' t \cdot \cos\varepsilon - \sin\omega' t \cdot \sin\varepsilon) -$$

$$-2\gamma A\omega' (\sin\omega' t \cos\varepsilon + \cos\omega' t \cdot \sin\varepsilon) +$$

$$+ \omega^2 A (\cos\omega' t \cdot \cos\varepsilon - \sin\omega t \cdot \sin\varepsilon) =$$

$$= \frac{F_0}{m} \cdot \cos\omega' t$$

$$A(\omega^2 - \omega'^2) \cos\omega' T \cdot \cos\varepsilon - 2\gamma A\omega' \cos\omega' T \cdot \sin\varepsilon -$$

$$- A(\omega^2 - \omega'^2) \sin\omega t \cdot \sin\varepsilon - 2\gamma A\omega' \sin\omega' t \cdot \cos\varepsilon =$$

$$= \frac{F_0}{m} \cdot \cos\omega' t$$

$$A(\omega^2 - \omega'^2) \cos\varepsilon - 2\gamma A\omega' \sin\varepsilon = \frac{F_0}{m}$$

$$-A(\omega^2 - \omega'^2) \sin\varepsilon - 2\gamma A\omega' \cos\varepsilon = 0$$

$$\Rightarrow A^2(\omega^2 - \omega'^2)^2 + 4\gamma^2 A^2 \omega'^2 = \left(\frac{F_0}{m}\right)^2$$

$$\Rightarrow A = \left(\frac{F_0}{m}\right) \cdot \frac{1}{\sqrt{(\omega^2 - \omega'^2)^2 + 4\gamma^2\omega'^2}}$$

1) $\omega' \ll \omega \Rightarrow A = \frac{F_0}{m\omega^2}$

2) $\omega' \gg \omega \Rightarrow A = \frac{F_0}{m\omega'^2}$

3) $\omega' = \omega \Rightarrow A = \frac{F_0}{m} \cdot \frac{1}{2\gamma\omega'}$

$\Rightarrow A = f(\omega)$

$$\frac{dA}{d\omega'} = 0 \Rightarrow A = A_{max}$$

Tendințe matematice

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$\omega^2 = \frac{l}{m}$$

# Curs 6

## Suprapunerea oscilațiilor

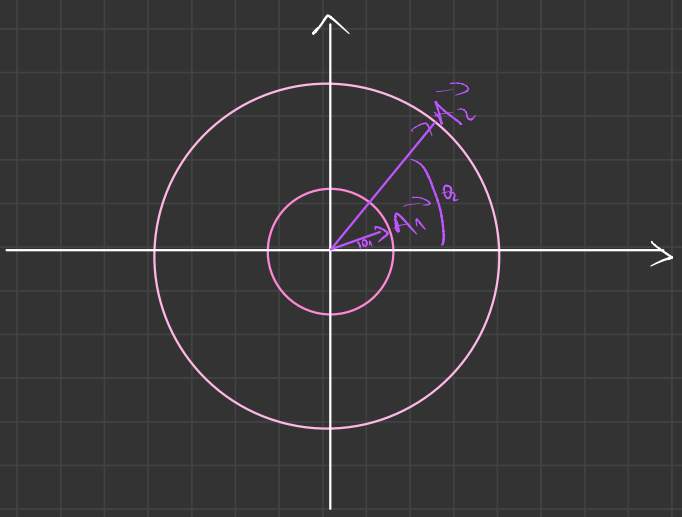
$$y = A\cos(\omega t + \theta_0)$$

Caz I: Suprapunerea oscilațiilor paralele

Ox: [resort] → $F_1 = -k_1 x \rightarrow x_1 = A_1\cos(\omega_1 t + \theta_{01})$

$F_2 = -k_2 x \rightarrow x_2 = A_2\cos(\omega_2 t + \theta_{02})$

## Reprezentarea fazorială a oscilațiilor




$$x = A\cos(\omega t + \theta_0)$$

$$x_R = A_R \cdot f(\omega t, \theta_0)$$

**Amplitudinea:** $A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos(\theta_{02} - \theta_{01})$

**Faza:** $\operatorname{tg}\theta_{02} = \dfrac{A_{1y} + A_{2y}}{A_{1x} + A_{2x}} = \dfrac{A_1\sin\theta_{01} + A_2\sin\theta_{02}}{A_1\cos\theta_{01} + A_2\cos\theta_{02}}$

Dacă $\begin{cases} A_1 = A_2 \\ \theta_{01} = \theta_{02} \end{cases}$ $\Rightarrow A\sqrt{2}$

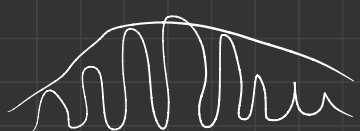
$\begin{cases} x_1 = A\cos(\omega_1 t) \\ x_2 = A\cos(\omega_2 t) \end{cases}$ $\Rightarrow x_2 = A(\cos(\omega_1 t) + \cos(\omega_2 t))$

$= 2A\cos\Omega t \cdot \cos\omega t,$

unde $\boxed{\Omega = \frac{\omega_1 t + \omega_2 t}{2}}$

$\boxed{\omega = \frac{\omega_1 t - \omega_2 t}{2}}$

$\Rightarrow x_{12} = 2A\cos\omega t \cos\Omega t$



efectul de BĂTĂI

Caz II: Oscilațiile sunt perpendiculare

$\begin{cases} x = A_1\cos(\omega_1 t) \\ y = A_2\cos(\omega_2 t) \end{cases}$

- $\theta_0 = 0$
  $\omega_1 = \omega_2$ / $\Rightarrow \begin{cases} x = A_1 \cos(\omega t) \\ y = A_2 \cos(\omega t) \end{cases}$ /

  $\Rightarrow y = \frac{A_2}{A_1} x$ (droite)

- $\theta_0 = \pi$
  $\omega_1 = \omega_2$ / $\Rightarrow \begin{cases} x = A_1 \cos(\omega t) \\ y = A_2 \cos(\omega t) \end{cases}$ /

  $\Rightarrow y = -\frac{A_2}{A_1} x$ (droite)

- $\theta_0 = \frac{\pi}{2}$
  $\omega_1 = \omega_2$ / $\Rightarrow \begin{cases} x = A_1 \cos(\omega t) \\ y = A_2 \sin(\omega t) \end{cases}$ /

  $\Rightarrow x^2 = A_1^2 \cos^2(\omega t)$
  $y^2 = A_2^2 \sin^2(\omega t)$ $(A_1 = A_2)$

  $\Rightarrow x^2 + y^2 = A^2$ (cercle)

- $\theta_0 \neq 0, \pi, \frac{\pi}{2}$
  $A_1 \neq A_2$ / $\Rightarrow$ ellipse

  $\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = 1$

# Analiza Fourier

$f(t) = f(t+T)$ , $T$ - perioada

$$f(t) = \sum_{n=0}^{\infty} A_n \sin(n\omega t + \alpha_n) =$$

$$= \sum_{n=0}^{\infty} A_n \left[ \cos(n\omega t)\cos\alpha_n - \sin(n\omega t)\sin\alpha_n \right]$$

$$= \sum_{n=0}^{\infty} a_n \cos(n\omega t) + b_n \sin(n\omega t) \quad / \cdot \cos(k\omega t)$$

$$\begin{cases} a_n = A_n \cos\alpha_n \\ b_n = A_n \sin\alpha_n \end{cases}$$

$\int_0^T$

$$\Rightarrow \int_0^T f(t)\cos(k\omega t)\,dt = \sum_{n=0}^{\infty} \int_0^T a_n \cos(n\omega t)\cos(k\omega t)\,dt +$$

$$\underbrace{\sum_{n=0}^{\infty} \int_0^T b_n \sin(n\omega t)\cos(k\omega t)\,dt}_{=0}$$

$$\delta_{nk} = \begin{cases} 0, & n = k \\ 1, & n \neq k \end{cases}$$

$$\Rightarrow \int_0^T f(t)\cos(k\omega t)\,dt = a_k \cdot \frac{T}{2}$$

$$\Rightarrow \boxed{a_k = \frac{2}{T}\int_0^T f(t)\cos(k\omega t)\,dt}$$

Analog $\int_0^T f(t)\sin(k\omega t)\,dt = b_k \cdot \frac{T}{2} \Rightarrow$

Deem: / $\cdot \sin(k\omega t)$

$$\int_0^T$$

$$\Rightarrow \boxed{b_k = \frac{2}{T}\int_0^T f(t)\sin(k\omega t)\,dt}$$

$$\int \sin(k\omega t) = \frac{e^{ik\omega t} + e^{-ik\omega t}}{2}$$

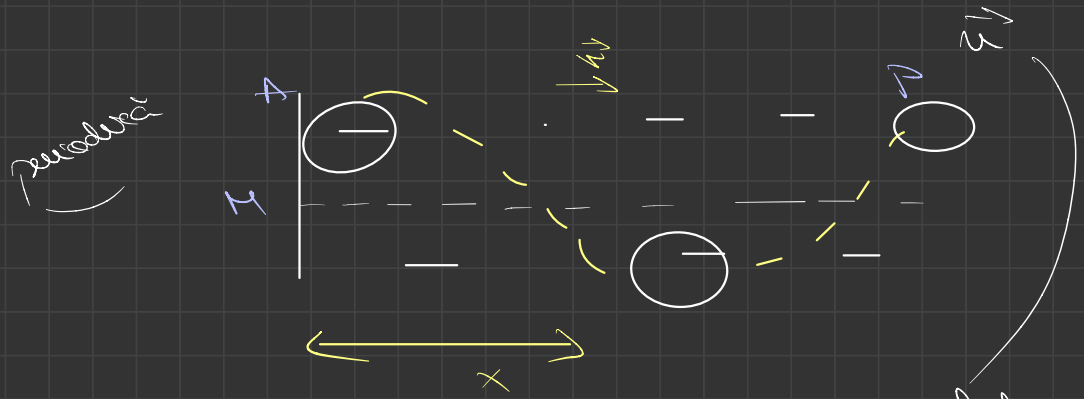
$$\cos(k\omega t) = \frac{e^{ik\omega t} - e^{-ik\omega t}}{2i}$$

$$\Rightarrow \boxed{f(t) = \sum_0^\infty c_n e^{i\omega t}}$$

$$\boxed{c_n = \frac{1}{T}\int_0^1 f(t) e^{i\omega t}\,dt}$$

Funcții neperiodice: $f(t) = \int_{-\infty}^{\infty} ccw \cdot e^{i\omega t}\,dt$

# Unda mecanică




$$v = \frac{x}{\Delta t}$$

$$\xi_x = A\cos(\omega t + \varphi_0)$$

$$\xi_M = A\cos(\omega(t - \Delta t) + \varphi_0)$$

$$\xi_M = A\cos\left(\omega t - \frac{\omega x}{v} + \varphi_0\right)$$

$$\begin{cases} \omega = \frac{2\pi}{T} \\ v = \frac{\lambda}{T} \end{cases}$$

$$\frac{\omega x}{v} = \frac{2\pi}{T} \cdot \frac{x}{\lambda} = \frac{2\pi}{\lambda} x \quad \Rightarrow \quad \boxed{\frac{2\pi}{\lambda} = k}$$

numărul de undă

$k$ = vectorul de undă, $\vec{k} = \frac{2\pi}{\lambda}\vec{n}$

$\varphi = A\cos(\omega t - kx + \theta_0)$

Funcția undă $\left\{\begin{array}{l} \text{Ec oscilator atunci când} \\ \text{un punct al oscilației} \end{array}\right.$

$y(x,t) = A\cos(\omega t - kx + \theta_0)$ - fct de undă

Variația în timp a lui $y$

$\frac{dy}{dt} = v_y$ - pulsația deci coardă

$\downarrow$ viteză oscilație ($v_y$)

$\rightarrow \frac{dy}{dt} = -\omega A\sin(\omega t - kx + \theta_0)$

$\rightarrow \frac{d^2y}{dt^2} = a_y = -\omega^2 A\cos(\omega t - kx + \theta_0) = -\omega^2 y$

$\downarrow$ acc oscilator ($\bar{a}_y$)

# Variația în spațiu a lui y

$$\frac{\partial y}{\partial x} = -kA\sin(\omega t - kx + \theta_0)$$

$$\frac{\partial^2 y}{\partial x^2} = -k^2 A\cos(\omega t - kx + \theta_0) = -k^2 y$$

$$\frac{1}{\omega^2}\frac{\partial^2 y}{\partial t^2} = \frac{1}{k^2}\frac{\partial^2 y}{\partial x^2} \Leftrightarrow \frac{\partial^2 y}{\partial t^2} = \frac{\left(\frac{\omega}{k}\right)^2}{v^2}\cdot\frac{\partial^2 y}{\partial x^2}$$

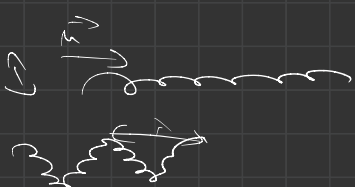
$$\boxed{\frac{\partial^2 y}{\partial t^2} = v^2\frac{\partial^2 y}{\partial x^2}}$$

Ecuația de propagare a undei $(x,t)$

Unde longitudinale și transversale

$f(x,t) = x(t)$

$x \parallel \vec{v}$




undă longitudinală oscilație $\parallel \vec{v}$

undă transversală oscilație $\perp \vec{v}$

# Cours 4

$y(x,t) = A\cos(\omega t - kx + \varphi)$ onda progresiva

$\vec{k}$, $x$ paralele și de același sens

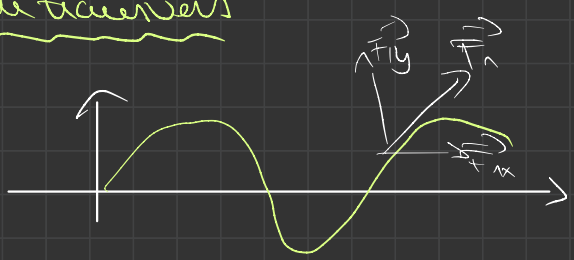
$\vec{k}$ ↑↑ $\vec{x}$ → undă regresivă

$y(x,t) = A\cos(\omega t - \vec{k}\vec{x} + \varphi)$

$\vec{k}\vec{x} = -kx \rightarrow y(x,t) = A\cos(\omega t + kx + \varphi)$

undă regresivă

## Undă transversală




$|F_{1x}| = |F_{2x}| = F$

$tg\alpha = \frac{F_{1y}}{F_{1x}} = \frac{F_{1y}}{F} = -\frac{F_{2y}}{F}$

$\left(\frac{\partial y}{\partial x}\right)_{x+\Delta x}$ $\left(\frac{\partial y}{\partial x}\right)_x$

$$F_{1y} = F\left(\frac{\partial y}{\partial x}\right)_{x+\partial x}$$

$$F_{2y} = -F\left(\frac{\partial y}{\partial x}\right)_{x}$$

$$F_y = F_{1y} + F_{2y}$$

$$\boxed{F_y = F\left(\frac{\partial y}{\partial x}\right)_{x+\partial x} + F\left(\frac{\partial y}{\partial x}\right)_{x}}$$

$$F_y = dm \cdot a = dm\left(\frac{\partial^2 y}{\partial t^2}\right) = \mu \partial x \left(\frac{\partial^2 y}{\partial t^2}\right)$$

$dm$ → masa lineal

$$dm = \mu \partial x$$

$$\frac{-F\left(\frac{\partial y}{\partial x}\right)_x + F\left(\frac{\partial y}{\partial x}\right)_{x+\partial x}}{\partial x} = \mu\left(\frac{\partial^2 y}{\partial t^2}\right)$$

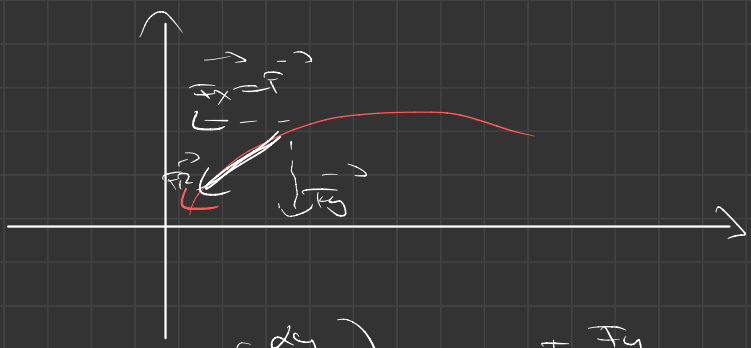
$\partial x \to dx$ ; $\frac{\partial^2 y}{\partial x^2}$

$$F\frac{\partial^2 y}{\partial x^2} = \mu \frac{\partial^2 y}{\partial t^2}$$

$$F \cdot \frac{1}{k^2} \cdot \frac{\partial^2 y}{\partial t^2} = \mu \cdot \frac{\partial^2 y}{\partial t^2} \Rightarrow \frac{1}{k^2} = \frac{\mu}{F}$$

$$\boxed{v^2 = \frac{F}{\mu}}$$

# Energía ondas



$$tg\alpha = \left(\frac{\partial y}{\partial x}\right)_x = \frac{+F_y}{F}$$

Potencia mecánica

$$P = \frac{\partial L}{\partial t} = \frac{\vec{F}\,\partial \vec{x}}{\partial t} = \vec{F}\vec{v} = -F_y \frac{\partial y}{\partial t}$$

$$F_y = F\left(\frac{\partial y}{\partial x}\right)_x$$

$$P = -F\left(\frac{\partial y}{\partial x}\right)\left(\frac{\partial y}{\partial t}\right)$$

$$y = A\cos(\omega t - kx + \varepsilon)$$

$$\frac{\partial y}{\partial x} = A \cdot k \cdot \sin(\omega t - kx + \varepsilon)$$

$$\frac{\partial y}{\partial t} = -A\omega \sin(\omega t - kx + \varepsilon)$$

$$P = -T \cdot A^2 \cdot 2 \cdot \omega \cdot \sin^2(\omega t - 2x + \varphi)$$

piedi campo

$$P = TA^2 \sin \cdot 2\omega \sin^2(\omega t - kx + \varphi)$$

$$P_{med} = TA^2 k\omega \qquad \langle \sin^2(\omega t + kx + \varphi) \rangle$$

$$P_{med} = \frac{T \cdot A^2 \cdot k\omega}{2}$$

$$\frac{\omega}{k} = v$$

$$P_{med} = \frac{TA^2\omega^2}{2v} =$$

$$= \frac{TA^2\omega^2}{2\sqrt{\frac{T}{\mu}}}$$

$$P_{med} = \frac{\sqrt{T\mu}\, A^2\omega^2}{2}$$

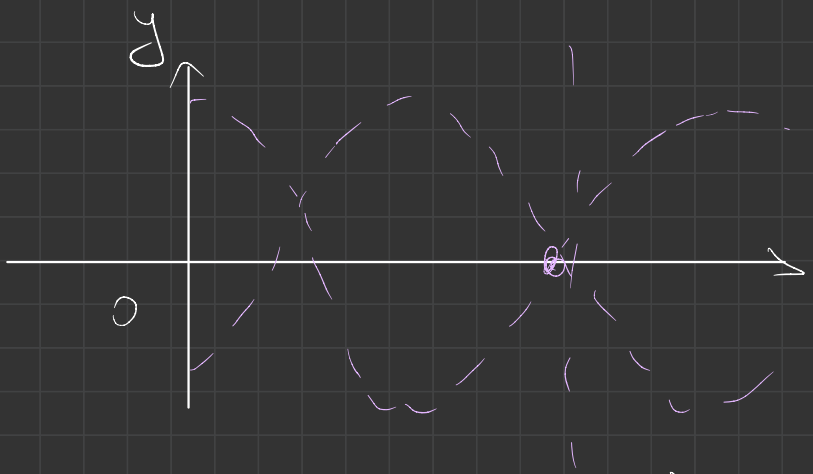
intensità onda:

$$I = \frac{P}{S}$$

$$I = \frac{P}{4\pi R^2}$$

onda sferica

# Onde stationnaire




$$y_1 = A\cos(\omega t - kx + \varphi)$$

$$y_2 = -A\cos(\omega t + kx + \varphi)$$

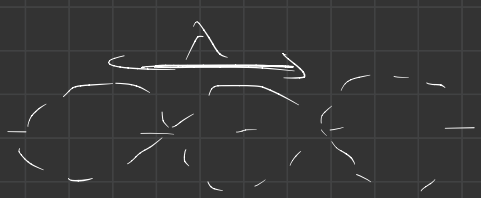
$$k = \frac{2\pi}{\lambda} \cdot \omega = \frac{2\pi}{\lambda} \cdot \nu$$

$$y = y_1 + y_2$$

$$A\left[\cos(\omega t - kx + \varphi) - \cos(\omega t + kx + \varphi)\right] =$$

$$= A\left[-2\sin(\omega t + \varphi)\sin(-kx)\right]$$

$$y = 2A\sin(kx)\sin(\omega t + \varphi)$$



## Posição nodais

$y = 0 \quad , \forall t \quad \Rightarrow \text{sen } kx = 0$

$kx = n\pi \Rightarrow$

$\frac{2\pi}{\lambda} x = n\pi \Rightarrow x = \frac{n\lambda}{2}$

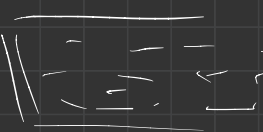
Ventre:

$\text{sen } kx = 1 \Rightarrow \frac{2\pi}{\lambda} x = \frac{\pi}{2}(2n+1)$

$\Rightarrow x = (2n+1)\frac{\lambda}{4}$

## Ondas longitudinais

Onda sonora

$\hookrightarrow$ 

$p(x,t) = -B\left(\frac{\partial y}{\partial x}\right) = -B \frac{\partial}{\partial x} A\cos(\omega t - kx + \phi)$

$= -B \cdot \text{sen}(\omega t - kx + \phi) A \cdot k$

$P_{max} = B \cdot k \cdot A$

## Viteza undelor longitudinale

$$v_t = \sqrt{\frac{T}{\mu}} \xrightarrow{\text{analog}} v_l = \sqrt{\frac{B}{\rho}}$$

unde $B$ modul elast volum

$\rho$ densitate gaz

## Intensitatea und. long.

$$I = \frac{P}{S} = p(x,t) \cdot v(x,t) = BAR \sin(\omega t - kx + \varphi) \cdot A\omega \sin(\omega t - kx + \varphi)$$

$$I = BA^2 k\omega \sin^2(\omega t - kx + \varphi)$$

$$\boxed{I_{max} = BA^2 k\omega}$$

## Nivelul sonor

$$\beta = 10 \log_{10} \frac{I}{I_0^2}$$

$$I_0 = 10^{-12} \ W/m^2$$

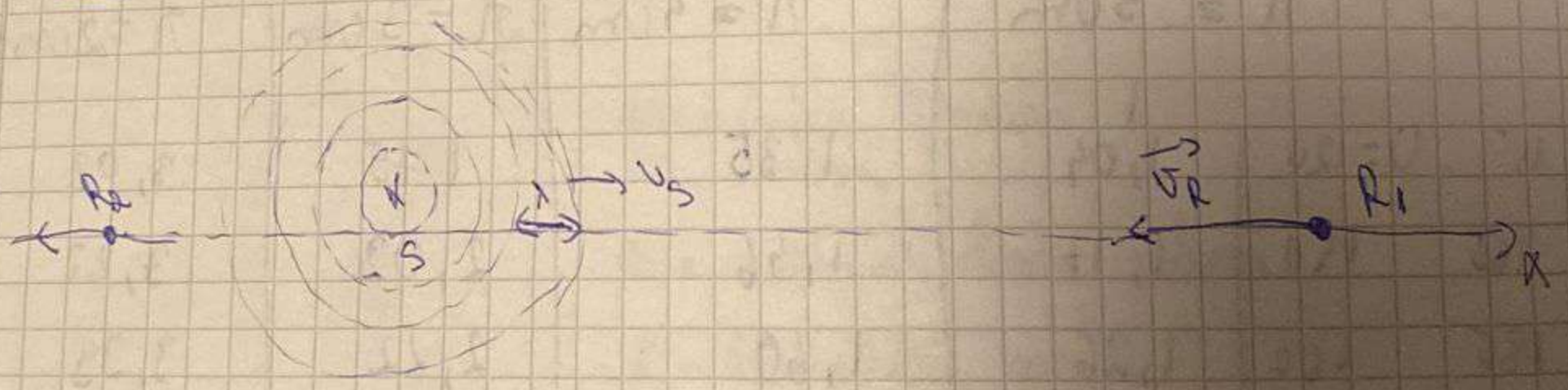
Curs fizica

Fizica modernă → nu la particul
semnată ca o particul
model de subiect - modelle

## Efectul Doppler

→ modificarea frecvenței unei unde sonore atunci când se deplasează receptorul / sursa

1. Sursa în repaus, receptorul în mișcare




$\lambda = v_s \cdot T = \frac{v_s}{\nu_s}$ ⇒ viteza sunetului

viteza undelor percepută de receptor:

$\vec{v} = \vec{v}_S - \vec{v}_R$

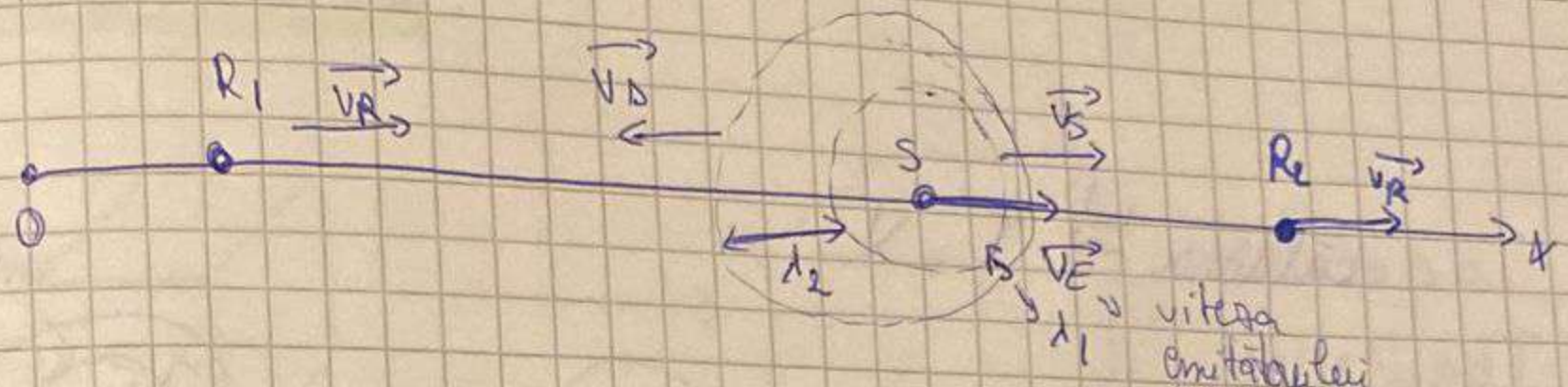
$v = v_S + v_R$

$\nu_R = \frac{v}{\lambda} = \frac{|\vec{v}_S - v_R|}{\frac{v_S}{\nu_S}} \Rightarrow \boxed{\nu_R = \nu_S \frac{|\vec{v}_S - \vec{v}_R|}{v_S}}$

$R_2$ se îndepărtează: $\nu_R = \nu_S \left(1 - \frac{v_R}{v_S}\right)$

$R_1$ se apropie: $\nu_R = \nu_S \left(1 + \frac{v_R}{v_S}\right)$

## 2. Sursa în mișcare




dist parcursă de frontul de undă în T este $d = v_s \cdot T$

distanța parcursă de sursă în T: $d' = v_E \cdot T$

lungimea de undă: $\lambda_1 = d - d' = (v_s - v_E)T = \dfrac{(v_s - v_E)}{\nu_s}$

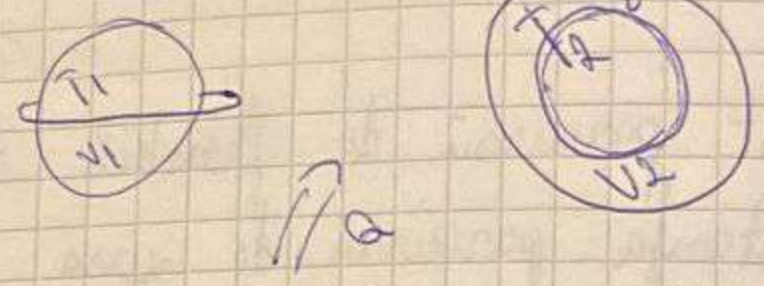
$$\lambda_2 = d + d' = \frac{(v_s + v_E)}{\nu_s}$$

$$\nu_{R_1} = \frac{|\vec{v} - \vec{v_R}|}{\lambda_2} = \frac{|\vec{v_s} - \vec{v_R}|}{v_s + v_E} \cdot \nu_s = \frac{v_R + v_s}{v_s + v_E} \cdot \nu_s$$

$$\nu_{R_2} = \frac{|\vec{v_s} - \vec{v_R}|}{\lambda_1} = \frac{v_s - v_R}{v_s - v_E} \cdot \nu_s \Rightarrow \boxed{\nu_R = \nu_s \cdot \frac{v_s \pm v_R}{v_s \pm v_E}}$$

# Fenomenele termice

căldura ⇒ T, V



$Q = \Delta U + L$

↳ var en interne

Q - primește energie

Cazul transformărilor infinitezimale : $\begin{cases} Q \to \delta Q \\ \Delta U \to dU = c \cdot dT \\ L \to \delta L = pdV \end{cases}$

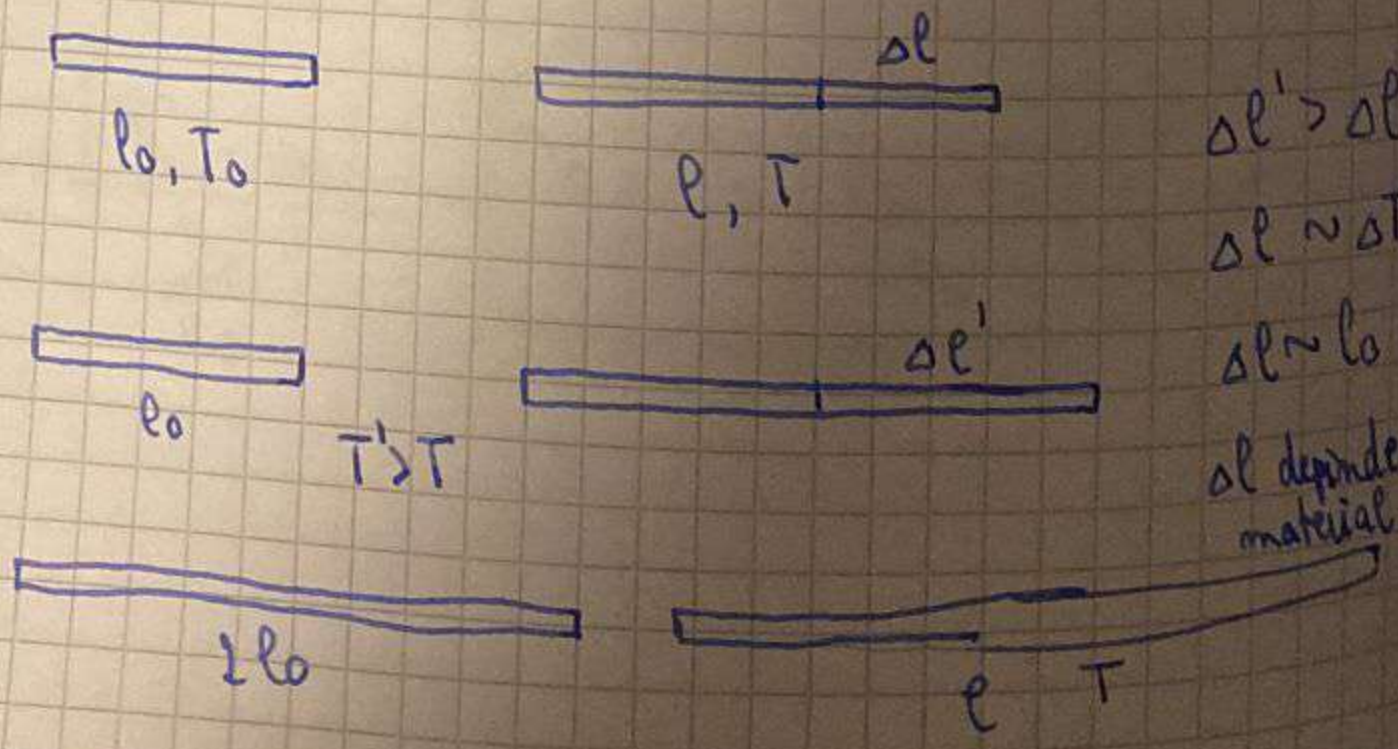
$\delta Q = dU + pdV$

$\delta Q = cdT + pdV$

Dilatarea termică : $\Delta V = f(\Delta T)$

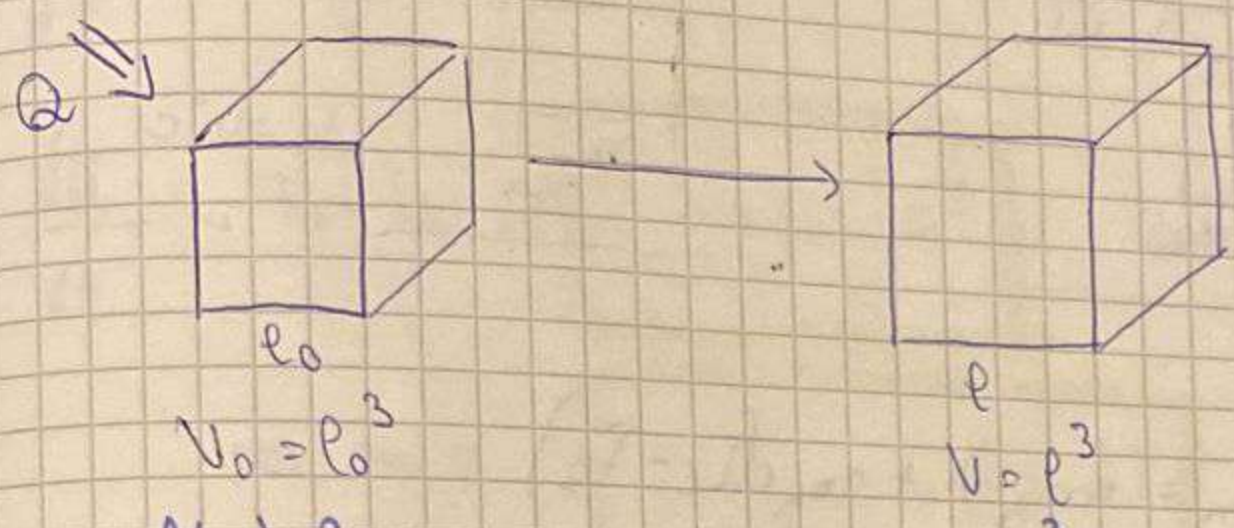
1. Cazul liniar

$\Delta l = f(\Delta T)$



$\Delta l = \alpha \cdot l_0 \Delta T$ —(în infinit zimale)→ $dl = \alpha l_0 dT$

## Dilatarea volumică




$\Delta V = V_0 \beta \Delta T$

$dV = V_0 \beta dT$

$\beta = f(\alpha)$ ?

$V = l^3 \rightarrow dV = 3 l_0^2 dl$

$V_0 \rightarrow V_0 + dV$

$3 \cdot l_0^2 \cdot dl = V_0 \cdot \beta \cdot dT$

$3 l_0^2 dl = l_0^3 \beta dT$

$3 \alpha l_0 dT = l_0 \beta dT$

$\beta = 3\alpha$

↑ coef de dilat liniară

↳ coef de dilatare volumică

## Căldura

moduri de exprimare

$\begin{cases} Q = mc\Delta T \\ dQ = mc\,dT \leftarrow \text{pt variații mici} \end{cases}$ c = căldura specifică

$\begin{cases} Q = C\Delta T \\ dQ = C\,dT \end{cases}$ C = capacitate calorică

$\begin{cases} Q = \nu C_0 \Delta T \\ dQ = \nu C_0 dT \end{cases}$ $C_0$ = căldura molară

$Q = m\lambda$ — $\lambda$ - căldura latentă (specifică) $\rightarrow dQ = \lambda\, dm$

## Transferul de căldură

① După atingerea echilibrului termic

$$Q_{primit} = m_g \cdot \lambda + m_g c (T_c - T_6)$$

$$Q_{ced} = -m_c \cdot c_c (T_c - T)$$

↑ căldură specifică

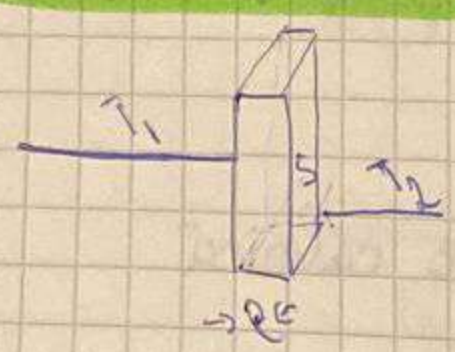
$$Q_{primit} + Q_{ced} = 0$$

$$\sum_i^n \delta Q_i = 0$$

② Fără atingerea echilibrului termic

- contact termic (conducție)
- radiație termică
- convecție

## Conducția termică




curentul de conducție termică

$$H = \frac{Q}{\Delta t} = \frac{dQ}{dt}$$

↑ flux termic

$H = \frac{\delta Q}{\delta t} \sim S$

$H = \frac{\delta Q}{\delta t} \sim \frac{1}{\ell}$

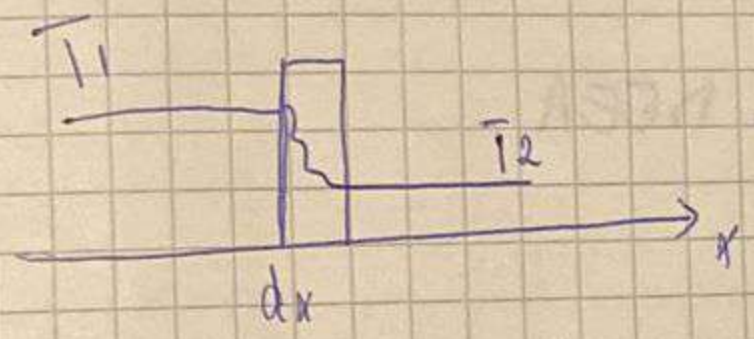
$H_2 \frac{\delta Q}{\delta t} \sim T_1 - T_2$

$H = \frac{k \cdot S}{L}(T_1 - T_2) = \frac{Q}{\Delta t}$

$H = \frac{-kS}{L} \cdot \Delta T$ $\quad \Delta T = T_2 - T_1$

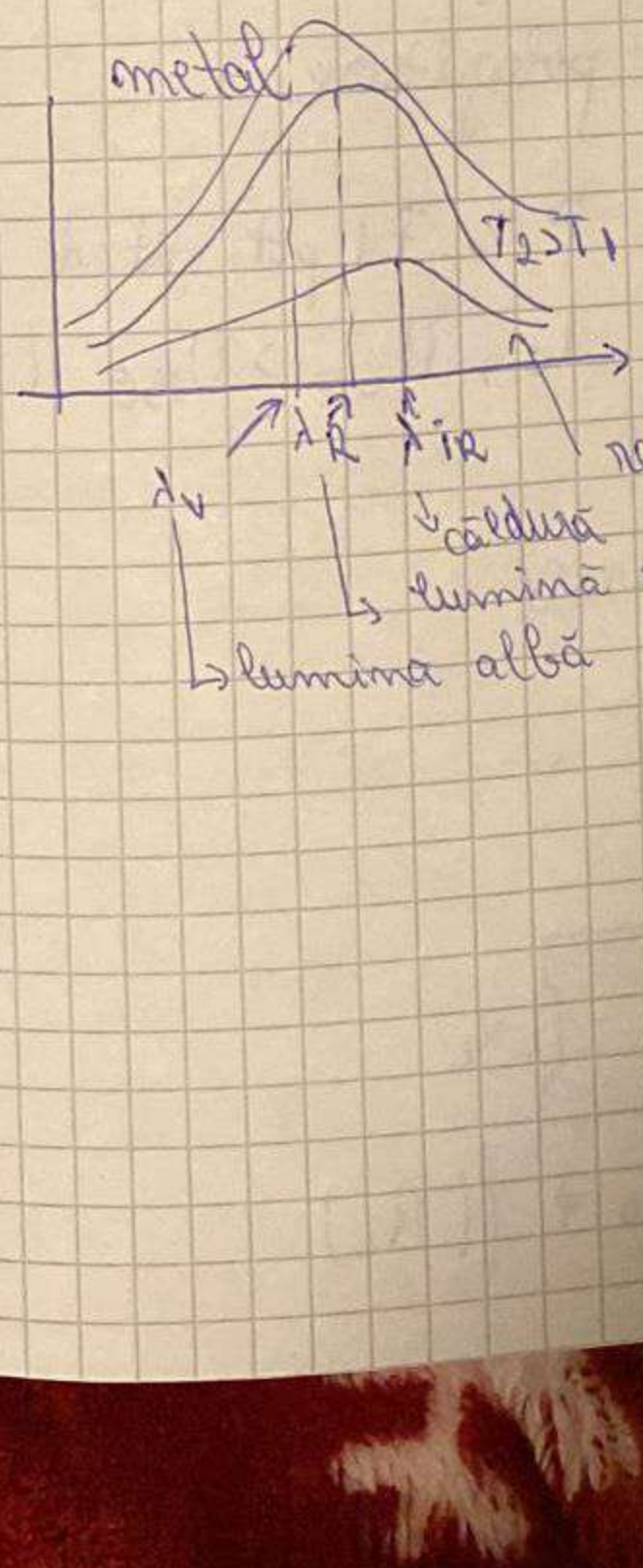
van mici $\quad \frac{\delta Q}{dt} = -\frac{kS}{\ell} dT$

→ când temperatura T nu scade uniform




$\Rightarrow \frac{\delta Q}{dt} = -kS\frac{dT}{dx}$

## Radiația termică




$H = \frac{dQ}{dt} = S \cdot e \cdot \sigma T^4$

S - aria corpului

e - emisivitatea, $e \in [0, 1)$

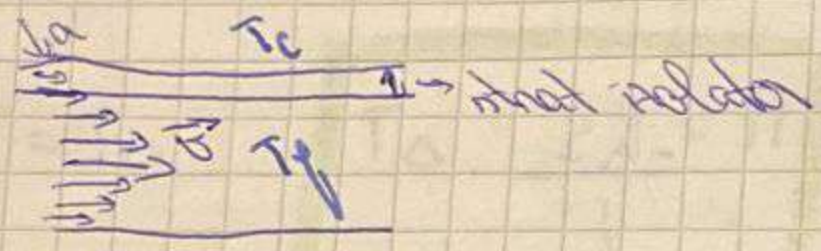
$\sigma$ = ct Stefan - Boltzmann

$\sigma = 5{,}67 \cdot 10^{-8}\ W/m^2K^4$

## Convectia

Obs experimentale

1. curentul de convectie este direct proportional cu S

2. 


convectie forțata, creștem viteza

3. $H \sim |\Delta T|^{5/4}$

curent de cond

$\Delta T = T_f - T_c$

fluid corp

Curs DEEA

RAN

BE pol directă

CB pol indirectă

$I_C = \beta I_B$

Componente

polarizare

P.S.F (pct stat de funcționare)

$I_C, U_{CE} > U_{BE}$ (MPN)

Stabilizatoa de tensiune

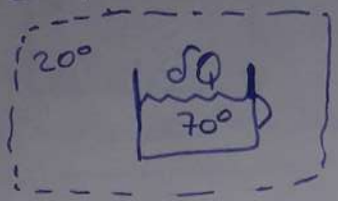
serie

15 · XI · 2022

# Curs 9 – Fizică

## Principiul al II-lea al termodinamicii

$$\delta Q = dU + p\,dV$$

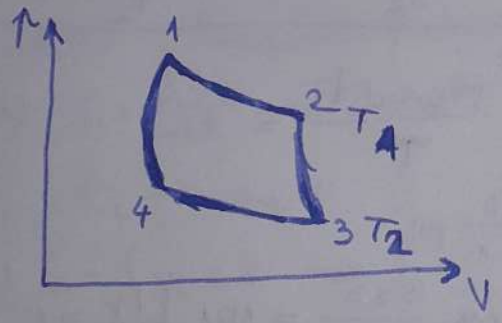



Formularea Clausius

Formularea Thomson

Randamentul: $\eta = \frac{L}{Q_1}$ ↓ $Q_1$ ↑

Randamentul maxim → ciclul CARNOT




1→2, 3→4 } tr. izoterme → $pV = ct$

2→3, 4→1 } tr. adiabate → $pV^{\gamma} = ct$

$$\eta_m = 1 - \frac{T_2}{T_1}$$

$$\eta = \frac{L}{Q_1} = 1 - \frac{|Q_2|}{Q_1}$$

$$Q_1 = L + |Q_2| \Rightarrow L = Q_1 - |Q_2|$$

$$1 - \frac{|Q_2|}{Q_1} = 1 - \frac{T_2}{T_1} \Rightarrow -\frac{|Q_2|}{T_2} + \frac{Q_1}{T_1} = 0 = \frac{Q_2}{T_2} + \frac{Q_1}{T_1}$$ (■)

tr. ciclică $\sum \frac{Q_i}{T_i} = 0$

transformări infinitezimale $\sum \frac{\delta Q_i}{T_i} = 0$

$\boxed{dS = \frac{\delta Q}{T}}$ ENTROPIA

$\boxed{\oint dS = \oint \frac{\delta Q}{T} = 0}$

## Entropia proceselor ireversibile

Exemplu:

$m_1 = 1\,kg\ H_2O \quad T_1 = 100°C \quad c = 4190\ J/kg \cdot K$

$m_2 = 1\,kg\ H_2O \quad T_2 = 0°C$

$\Delta S = ?$

cond. de ech.: $m_1 c_1 (T_1 - T_e) = m_2 c_2 (T_e - T_2)$

$m_1 c_1 = m_2 c_2$

$\Rightarrow T_e = \frac{T_1 + T_2}{2} = 50°C$

$\left.\begin{aligned} dS &= \frac{\delta Q}{T} \\ \delta Q &= mc\,dT \end{aligned}\right\} \Rightarrow dS = \frac{mc\,dT}{T} \Rightarrow \boxed{\Delta S = \int_{S_i}^{S_e} dS = S_e - S_i = \int_{T_i}^{T_e} \frac{mc\,dT}{T}}$

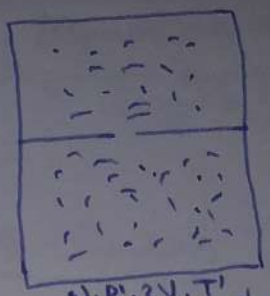
$\Delta S_t = \Delta S_1 + \Delta S_2 = \int_{T_1}^{T_e} \frac{m_1 c_1 dT}{T} + \int_{T_2}^{T_e} \frac{m_2 c_2 dT}{T} =$

$= m_1 c_1 \ln \frac{T_e}{T_1} + m_2 c_2 \ln \frac{T_e}{T_1}$ (*)

$\Delta S = mc \ln \frac{T_e^2}{T_1 T_2} = 1 \cdot 4190 \cdot \ln \frac{323^2}{373 \cdot 273} = 101\ J/K > 0$

## Interpretarea statistică a entropiei

$S = k_B \ln W$




$W' = 2^N W$

$S' = k_B \ln W'$

$W$ = nr. de microstări comparabile cu macrostarea dată

---

① $v_0 = 4\ m/s$

$a = -3v^2$

$t = ?$ $a \uparrow \cdot v = \frac{v_0}{2}$

$d = -3(m^{-1})$

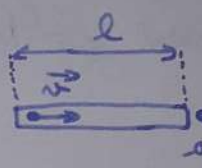
$a = \frac{dv}{dt} \Rightarrow \int_{v_0}^{v} dv = \int_0^t a\,dt$

$-3v^2 = \frac{dv}{dt} \Rightarrow \int_0^t dt = \int_{v_0}^{v} -\frac{1}{3v^2}\,dv = \frac{1}{3v}\Big|_{v_0}^{v} = \dots$

1.1.

② $v = 3 \cdot 10^5 t - 5 \cdot 10^7 t^2$ (m/s)

$a_f = 0$




$\tau = ?$ (timpul după care iese din pușcă)

$l = ?$ (lungimea țevii)

a) $a = \frac{dv}{dt} = 3 \cdot 10^5 - 10^8 t$

$t_f \Rightarrow a_f = 0$

$10^8 t_f = 3 \cdot 10^5 \Rightarrow t_f = 3\ ms$

b) $x = \int v\,dt \Rightarrow l = \int_0^{t_f} (3 \cdot 10^5 t - 5 \cdot 10^7 t^2)\,dt = \frac{3}{2} \cdot 10^5 t_f^2 - \frac{5 \cdot 10^7 \cdot t_f^3}{3}$

③ $m = 5\ kg$

$\vec{v_1} = 6\vec{i} - 2\vec{j}$

$\vec{v_2} = 8\vec{i} + 4\vec{j}$

$L = ?$

$L = \Delta E_c = \frac{mv_2^2}{2} - \frac{mv_1^2}{2} = \frac{5 \cdot 80}{2} - \frac{5 \cdot 40}{2} = 100J$

④ $R = 220\,m$

$v = 25\,m/s$

a) $\mu = ?$

b) gheață $\Rightarrow \mu' = \frac{\mu}{3}$

$v' = ?$

$m v^2 = \mu m g$

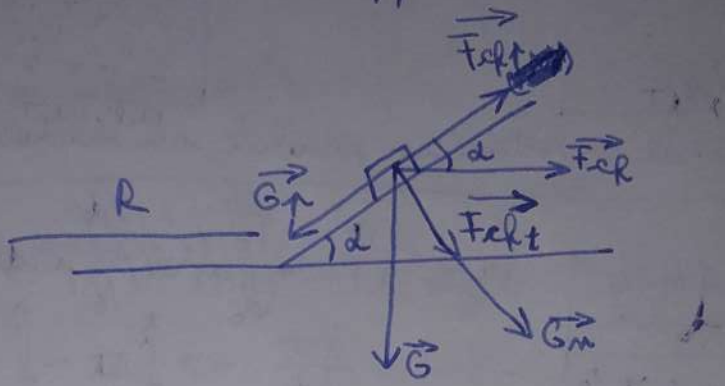
$\mu = \frac{v^2}{Rg}$

⑤ $m = 1125\,kg$

$R = 225\,m$

$v = 65\,mi/h$

$\alpha = ?$




$G_t = F_{cp\uparrow}$

$mg \sin\alpha = \frac{mv^2}{R} \cos\alpha \Rightarrow tg\alpha = \frac{v^2}{Rg}$

⑥ $m = 1000\,kg$

$\Delta x = 3{,}16\,cm$

$K = 5 \cdot 10^6\,N/m$

$v_i = ?$

$\Delta E_c = L = \frac{k \Delta x^2}{2}$

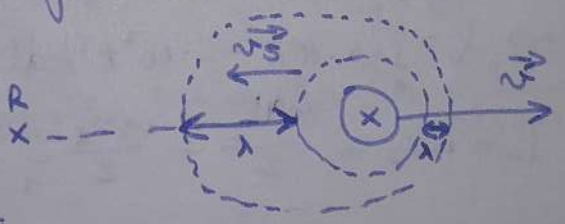
⑦ $\nu = 600\,Hz$

$H = 15\,m$

$v = 340\,m/s$

$\nu' = ?$

$mgh = \frac{mv^2}{2} \Rightarrow \boxed{v = \sqrt{2gh}}$




$\lambda = \frac{v_s + v}{\nu_s}$